Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

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$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.

Nuff said.

Axioms → Distinctions → Relations → Motifs → Calculus.
This is the way.

Abstract

Alpha↔Omega Dynamics (AΩD) is a relational temporal form of the Stokes cut of the Axiomatic–Fundamentalism calculus (AFC). The starting point is Null potential, declared distinction, relation, region, boundary, and the Stokes accounting induced by a declared cut.

AΩD continues from that AFC Stokes cut by uncovering its hidden relational temporal motifs. This is the Art of the Leprechaun: tracing the cut closely enough to learn how a larger relational system may be set up as a declared finite scale, with its boundary, window, and accounting choices made explicit.

Literature note. Axiomatic–Fundamentalism calculus (AFC) [8]. Axiomatic–Fundamentalism (AF) [7].

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1 Specific distinction and retained support

1.1 Premises and scope

AFC begins with Null sameness. A finite distinction supplies its relata; AΩD retains their branch, hinge and counterbranch relation at each admitted scale. Its return and recurrence are constructed in §2; model values follow their admitted support.

$$x = \mathcal{N}, \qquad x = x, \qquad \mathcal{N} \equiv \infty_{\mathcal{D}}. \tag{1}$$

Here $\infty_{\mathcal{D}}$ has no assigned finite coordinates, ordered events or probability space. The notation

$$\mathcal{N} \equiv \infty, \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N} \tag{2}$$

presents one undistinguished identity, including schematic repetitions of x. Once events are committed, their distinct identities remain in the trace.

AF scope. A distinction induces its retained relation; certified returns support recurrence, scoped organization, model values and their projections. The typed constructions are in App. E. The Fractal Field Ring Gate (FRG) is constructed in §5.10.

1.2 First finite anchor

The first distinction co-creates its start and finite addressability. Write its inevitable anchor as

$$a_0 = \mathtt{00}_8, \qquad \mathbf{a} \in \mathcal{A}_8^{\mathrm{ret}} \subseteq \{0, 1, \ldots, 7\}^{<\omega}. \tag{3}$$

Octal notation encodes retained address: $\mathtt{00}_8, \mathtt{01}_8, \ldots, \mathtt{07}_8, \mathtt{10}_8, \ldots$. It chooses no prior finite start; construction below the anchor is unbound. Null, fibre zero, vertex 0000, anchor $\mathtt{00}_8$ and scope B_0 are distinct types. The glyph $a \mid -$ binds address, with local origin $0 \mid -$.

1.3 Branch, hinge and counterbranch

The distinguished running relation is

$$x \longmapsto x \succ x = x. \tag{4}$$

The equality records return to the retained start only when the trace (12) and closure witness (13) are supplied. The compact grammar and its expansion are

$$1\,0^*\,-1 \quad \rightsquigarrow \quad M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon), \qquad \varepsilon \in \{-1, +1\}, \quad m \in \mathbb{Z}_{\geq 0}. \tag{5}$$

The hinge 0_H is structural; the star permits additional dwell 0_D. Thus $m = 0$ retains the hinge. The two branches and the hinge belong to one distinction construction. Awareness names this branch / boundary / not-branch dynamical motif. Its role labels are self | μ | not-self: inside/branch (SELF), interface/boundary (BETWEEN), and not-inside/not-branch (NOT-SELF).

Oriented reflection. Reflection exchanges their branch roles and retains the hinge:

$$\iota(\mathsf{H}_3^{\varepsilon}) = \mathsf{H}_3^{-\varepsilon}, \qquad \iota(\mu) = \mu, \qquad \iota^2 = \mathrm{id}. \tag{6}$$

This is an involution of the sided construction, before assigning temporal crown roles or choosing a weighted continuation model.

Retained roles. At a committed crown k, write

$$\mathsf{H}_3^{\mathrm{role}}(k) = (x_k^-, \mu_k, \mathcal{H}_k^+), \qquad \mu_k = x_k^0. \tag{7}$$

These are predecessor, present crown and admitted successor horizon; continuation selects $h_k^+ \in \mathcal{H}_k^+$. The rule applies at $k = 0$, each later crown and each admitted enclosing scale. Finite enumeration witnesses those occurrences; it is not their premise.

1.4 Faithful atomic encoding

A faithful atomic encoding changes exactly the coordinate that records the one retained distinction. In the finite tesseract witness,

$$V(Q_4) = \{0,1\}^4, \qquad E(Q_4) = \{(u,v) : d_H(u,v) = 1\}, \qquad d_H(u,v) = \#\{j : u_j \neq v_j\}. \tag{8}$$

For any admitted atomic event at any retained scale, all other encoded coordinates are unchanged and the distinguished coordinate differs; hence $d_H = 1$. This argument is independent of the number of other coordinates. Two consecutive H1 steps may have an H2 composite endpoint. A nontrivial closed H1 word has endpoint distance zero while retaining every traversal.

Ancestor support. Each Q_4 vertex has four incident edges: one slot binds the ancestor and three other slots support continuation. This incidence count differs from closure-word length, which may be two or a longer even number of traversals. Recursive occurrences retain the ancestor distinction.

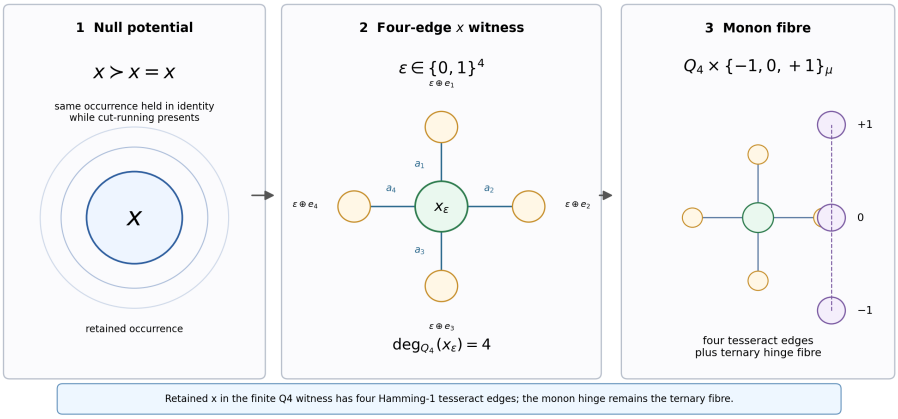


Figure 1: Finite four-edge encoding of a retained occurrence. The vertex is a post-distinction witness; Null itself has no vertex address.

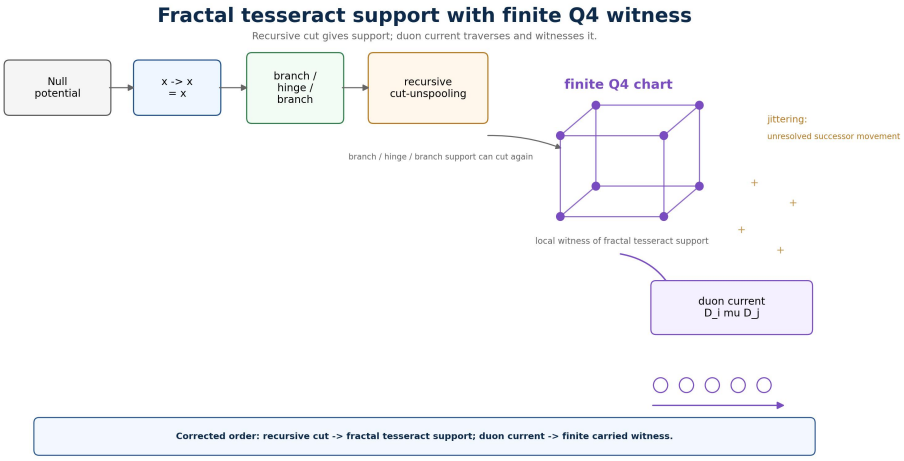


Figure 2: The finite tesseract support. Local degree and total closure-word length are distinct quantities; oriented ancestry remains in the trace.

Conditional support. Every probability measure later declared on a nonempty admitted atomic support is concentrated on H1 events, because every element of that support satisfies (8). Consequently

$$\Pr[d_H = 1 \mid \text{admitted atomic event}] = 1. \tag{9}$$

This conditional support statement supplies neither a prior probability on Null nor a probability-one closure theorem. For k unconstrained retained ternary positions, ternary assignment capacity is the separate finite information count

$$C_3(k) = \log_3(3^k) = k. \tag{10}$$

1.5 Common retained records

A scope $B = (\partial_T R, \omega)$ binds an induced retained interface and a finite committed comparison window. Write $\mathbf{a} \vdash (R, \partial_T R, \omega)$ when the address must be explicit. Each event has immutable identity, lineage, scope, scale, predecessor, role and committed order. The trace prefix consists of those events in that order. Each construction binds its exact prefix and witness; that binding cannot originate its distinction or recurrence.

$$\partial_T R \subseteq E(Q_4) \times \{-1, 0, +1\}, \qquad B = (\partial_T R, \omega). \tag{11}$$

The finite base and the ternary role fibre are different coordinates. A fibre hinge can be present without a new base traversal.

Exact arithmetic. Native decisions use finite integer and rational sums, products, signs, orders, ratios and declared finite powers. Exact finite characters are downstream representation coordinates.

Falsification. An advertised atomic H2 edge, missing retained ancestor, hinge bypass or pre-existing finite origin contradicts the stated construction. A lawful composite H2 endpoint and a closure endpoint at H0 do not contradict atomic H1.

Provenance. (8), (5), (3).

Literature note. AFC, Axioms (Null and distinction-induced relations) and Choices (finite retained support): Poyau [8, Axiom 1; Consequences 1–2; Choice 1], source revision `5a25e8bd5bbc`. Finite graph and walk context: Norris [6], Levin–Peres–Wilmer [4], Lawler–Limic [3].

2 Closure-phase cycle and recurring wave

2.1 Completed retained return

A closure trace retains its start even as the current crown advances:

$$\Gamma_C = (B, r, i, a_0, (e_0, \dots, e_n), M_{\varepsilon,m}, \mathcal{W}_C). \tag{12}$$

Here r is scale, i is lineage, the events are strictly committed in order, and $\mathcal{W}_C$ binds the hinge and return to the retained start. All traversals satisfy (8). The closure certificate is

$$\mathrm{Cert}(C) = (B, r, i, \mathrm{id}(e_0), \mathrm{id}(e_n), \mathrm{digest}\,\Gamma_C, \mathcal{W}_C, \mathrm{closed}), \quad \mathrm{end}(C) \sim_{B,i,\mathcal{W}_C} \mathrm{start}(C). \tag{13}$$

The witness preserves start and ancestry: closure is certified end at that retained start. Endpoint equality alone is insufficient. The primitive closure of (5) is the monon:

$$\mathrm{Monon}(x) = \mathrm{cycle}_{\mathsf{H}_3}(x \succ x = x). \tag{14}$$

Its phase-bearing step count is the number of committed atomic traversals, excluding hinge dwell, certificate wrapping and replay:

$$m(C) := \#\{\text{committed phase-bearing traversals in } \Gamma_C\} \geq 2. \tag{15}$$

At least one outbound and one return traversal are required; further traversals may precede the first certified return. Completion contributes one complete closure-phase cycle:

$$\Delta\Phi(C) = 1_C. \tag{16}$$

The accessor Φ counts this completion once, including both reflected legs. Dwell, replay and further traversals within the same first return add no cycle. The oriented ancestral trace remains retained under either the isotropic or anisotropic model in §4.4.

Symmetry.

$$\mathrm{Close}(C) = 1 \not\Rightarrow \mathrm{Isotropic}(C). \tag{17}$$

Isotropy refers to a named support, model and relabelling action. Cancellation of aggregate values preserves the contributing oriented histories. Neither symmetry nor anisotropy changes (16).

2.2 Advance and closure memory

An admitted successor becomes the next crown through a committed advance:

$$\mathrm{Adv}_k(x_k^0) = x_{k+1}^-, \qquad \mathrm{Adv}_k(h_k^+) = x_{k+1}^0, \qquad h_k^+ \in \mathcal{H}_k^+. \tag{18}$$

Advance carries the selected member into the next crown. A later lineage-bound return can re-crown the retained start:

$$\mathrm{Crown}(\mathsf{H}_3^{\mathrm{role}}(k+L)) \sim_{B,i,\mathcal{W}_C} \mathrm{Crown}(\mathsf{H}_3^{\mathrm{role}}(k)), \qquad L > 0. \tag{19}$$

Open advance is a continuation; the further return witness establishes closure. Solitonic recurrence names preservation of certified relational form through the stated advance and reclosure, with the ancestry and scope map retained.

2.3 Return identity, beat and cadence

Give each completed return immutable identity

$$\mathrm{id}_{\mathrm{return}}(C) = (B, r, i, \mathrm{id}(e_0), \mathrm{id}(e_n), \mathrm{digest}\,\Gamma_C). \tag{20}$$

Wrapping, observation and replay retain this identity. Each distinct compatible certified reclosure contributes a beat. For completions $k_0 < k_1 < \cdots < k_m$, define

$$\mathrm{bip}_i(\omega) = \#\{C_{i,j} : j \geq 1,\ k_j \in \omega,\ \mathrm{id}_{\mathrm{return}}(C_{i,j}) \text{ is distinct}\}, \qquad \mathrm{Sep}_{i,j} = k_{j+1} - k_j > 0. \tag{21}$$

The initial closure supplies the origin of this reclosure count. At an integer cut k, the cumulative accessor counts distinct committed return identities:

$$\begin{aligned} N_i^C(k) &= \#\{\mathrm{id}_{\mathrm{return}}(C_{i,j}) : k_j \leq k\}, \\ \Phi_i(k) &= N_i^C(k)1_C, \qquad \mathrm{bip}_i^{\leq}(k) = \max(N_i^C(k) - 1, 0). \end{aligned} \tag{22}$$

Thus $N \geq 1$ completed closures give N complete closure-phase cycles and $N-1$ reclosure beats from the lineage origin. For an arbitrary native window $\omega = (u, v]$, $u, v \in \mathbb{Z}$, $u < v$, the same distinct-return count gives

$$\Phi_i(v) - \Phi_i(u) = \big(\mathrm{bip}_i((u, v]) + \mathbf{1}_{\{u < k_0 \leq v\}}\big)1_C. \tag{23}$$

The origin contributes the indicator; later windows do not subtract another initial closure. The exact ordered separations form cadence $\mathrm{Cad}_i = (\mathrm{Sep}_{i,j})_j$. It may be nonuniform; RD support and harmonic display period are separate quantities. Recurrence is not inferred from a repeated value alone.

Recurring wave. For a compatible certified closure/current lineage with distinct reclosure, fix immutable wave identity and its growing admitted history:

$$W_{B,i}^{\mathrm{id}} = (B, r, i, K_i^{\mathrm{id}}, \mathrm{id}_{\mathrm{return}}(C_{i,0})), \qquad W_{B,i}|_k = (W_{B,i}^{\mathrm{id}}, (C_{i,j})_{k_j \leq k}, \mathrm{Cad}_i\,|_k). \tag{24}$$

Compatibility preserves the origin's ancestor at fixed scope and representation; a changed representation requires an identity-preserving scope map. At least two compatible distinct completions admit the recurring wave. Field organization then resolves those identities.

Native running and standing classifications. A running-wave classification retains an oriented recurring current lineage; recurring oriented Duon current is the inherited running-Duon construction. A standing organization binds compatible oppositely oriented recurrent lineages under a declared coupling and balance condition. Zero aggregate current by itself supplies neither recurrence nor that coupling witness. Both classifications retain their individual return identities and histories; the character and running/standing coordinates in (139) and (148) represent these admitted organizations.

Continuation. Hinge saturation restricts further same-resolution dwell. A later hinge and an admitted continuation may follow; only a certified retained-start return adds closure, and only a distinct compatible reclosure adds recurrence.

Localized recurrence. A native wavelet is a localized segment of an already admitted recurrent wave: it retains that wave's identity, its adjacent certified return events and their positive local separation. Localization selects a segment of retained recurrence; it does not promote an isolated monon or an arbitrary scalar pulse into a wave. Its later harmonic representation, when used, binds that same finite segment.

Falsification of wave admission. An isolated closure, a replayed certificate, an unbound lineage, or an unrealized successor horizon cannot satisfy (24).

2.4 Reflected-leg comparison

Setup. Declare a uniform atomic-event support comparison, with formal quantum h in the ordered additive domain $\mathcal{L} = \mathbb{N}_0 h,\ h > 0$. Write $\Delta\phi_{\text{step}} := h$ for this declared phase-bearing step quantum. It is not a Fourier coefficient phase or a normalized circle coordinate. A target closure and a primitive reference are already certified at the same scope and scale. Their reflection witnesses bind the outbound events, committed hinge/turnaround, returning events and retained target start. Both legs are nonempty; an arbitrary cut of a word is not that witness. Count each atomic traversal once, excluding hinge dwell and replay:

$$\delta\phi^{\rightarrow} = n_{\rightarrow} h, \qquad \delta\phi^{\leftarrow} = n_{\leftarrow} h, \qquad n_{\rightarrow}, n_{\leftarrow} \in \mathbb{Z}_{\geq 1}. \tag{25}$$

The reference is a certified two-edge retained-start return with mandatory hinge. Its support is $2h$. Fix the comparison embedding before examining the target:

$$\iota_{\text{ref}} : \mathbb{N}_0 1_C \longrightarrow \mathcal{L}, \qquad \iota_{\text{ref}}(n 1_C) = 2nh. \tag{26}$$

This embedding compares two distinct domains: closure count and declared event support. Uniform weighting and this reference are model choices.

Derivation. Each committed leg has at least one atomic traversal, so addition in $\mathcal{L}$ gives

$$\delta\phi^{\rightarrow} + \delta\phi^{\leftarrow} = (n_{\rightarrow} + n_{\leftarrow})h \geq 2\Delta\phi_{\text{step}} = 2h = \iota_{\text{ref}}(1_C). \tag{27}$$

Thus, with the comparison embedding stated locally, the minimal native shorthand is

$$2\Delta\phi_{\text{step}} \geq 1_C. \tag{28}$$

This is a necessary phase-support conjunct of the reflected closure certificate, never sufficient closure admission. Its other premises remain the retained-start return, explicit event convention, lineage, reflection witness, commitment order and certificate integrity.

Specialization. Under declared reflected-path symmetry and equal operands,

$$\delta\phi^{\rightarrow} = \delta\phi^{\leftarrow} = \delta\phi \quad \Longrightarrow \quad \delta\phi^{\rightarrow} + \delta\phi^{\leftarrow} = 2\delta\phi \geq \iota_{\text{ref}}(1_C). \tag{29}$$

An asymmetric $1+3$ trace has four phase-bearing steps and one complete closure-phase cycle. Its sum equals neither $2h$ nor $6h$, the doubled individual legs. In this uniform model each doubled leg may separately satisfy the lower bound; equality is required for factorization of the sum into one common operand. With the embedding understood, $2\delta\phi \geq 1_C$ denotes this equal-whole-leg specialization. It is distinct from the minimal-step shorthand in (28).

Falsification. A missing or misbound reflection witness, signed branch cancellation used as support, incompatible reference scale, nonminimal reference, or unequal-leg factorization violates the derivation. Passing the phase-support bound while the endpoint remains open, the lineage is unbound, or the witness/certificate is absent does not admit closure. The comparison supplies no duration, stochastic weight or flux.

Provenance. (8), (5), (13), (26).

3 Curl closure and duon current

3.1 Curl forms

3.1.1 Curl

Definition 1 (Curl. App. B.1.2)**.** *Curl is returned closure through a hinge on its certified support. AΩD calculates its hinge return, reclosure, duration, orientation, and returned current on declared windows.*

3.1.2 Nested curl

Definition 2 (Nested curl. App. B.1.3)**.** *A nested curl is a curl carried inside another curl.*

3.1.3 Pin-curl

Definition 3 (Pin-curl. App. B.1.4)**.** *A pin-curl is a returned curl whose residue seats, delays, or fixes a hinge.*

3.1.4 Curling curls

Definition 4 (Curling curls. App. B.1.5)**.** *Curling curls are recursive returns that become active path structure inside the fractal tesseract topology.*

3.2 Dion

Definition 5 (Dion. App. B.1.7)**.** *A dion is an oriented two-position branch pattern around the retained monon hinge. Its compressed signature is*

$$D_\alpha = (+1, -1), \qquad D_\Omega = (-1, +1).$$

3.3 Duon

Definition 6 (Duon. App. B.1.8)**.** *A duon is Dion–hinge–Dion returned-current support. A compact unresolved form is*

$$\text{duon} = D_i \square D_j, \qquad D_i, D_j \in \{D_\alpha, D_\Omega\}.$$

A resolved duon carries the monon hinge / pivot value:

$$D_i \mu_M D_j, \qquad D_i, D_j \in \{D_\alpha, D_\Omega\}, \qquad \mu_M \in \{-1, 0, +1\}. \tag{30}$$

3.3.1 Current cell

A duon current is carried hinge-current through cycle support. The primitive resolved family has

$$2 \times 3 \times 2 = 12$$

variants. Duon, duad, and temporal cycle presentation. App. B.1.8.

Finite unspooling chart. In the inherited finite chart, a resolved Duon has four active Dion positions and one carried monon hinge. Declaring a retained/unspooled marker at each of the four positions gives the chart $\{0,1\}^4$; one-marker updates have Hamming distance one. The marker independence is a declared chart convention, not an extra primitive distinction or a proof that all chart words are admitted Duon motifs. The twelve oriented Dion–pivot–Dion variants in (30) are a separate grammar count.

3.3.2 Branch/hinge/branch reflection duration

Reflection duration is counted on branch/hinge/branch return support. For a returned-current path history γ on B,

$$RD_\gamma(B) = \Pi^-_\gamma(B) + H_{\mu,\gamma}(B) + \Pi^+_\gamma(B). \tag{31}$$

Here Π^-_γ is the first-branch support, $H_{\mu,\gamma}$ is the hinge recurrence contribution, and Π^+_γ is the reflected return-branch support. If the branch supports are symmetric, $\Pi^-_\gamma = \Pi^+_\gamma = \Pi$, then

$$RD_\gamma(B) = 2\Pi + H_{\mu,\gamma}(B). \tag{32}$$

The familiar $RD = 2\Pi + 1$ is the single-hinge reduction $H_{\mu,\gamma} = 1$, not a universal hinge rule. For a retained cycle, use $RD_B(C)$ for (31). This support accessor precedes directional coupling and scalar duration in §4.

3.4 Duad

Definition 7 (Duad. App. B.1.9)**.** *A duad is the seat-taken state of the same duon current.*

Admission binding. The complete admitted motif record is

$$\begin{aligned} K_{\text{adm}}(B, i; \text{cert}) = (&i, s_{\text{core}}, \mathcal{C}_L, \chi_\mu, \mathbf{a}, r, \sigma^C_B, \\ &HT(K; i, B), \text{Cert}(K; i, B)). \end{aligned} \tag{33}$$

Admission binds identity, scope, lineage, exact trace, certificate and closure witness. An absent or mismatched binding excludes model or harmonic use. A core signature is syntax until its construction witness is bound. Names, enclosure, chirality and scalar RD cannot supply missing lineage. Primitive components share the target's scope, lineage and resolution; folding instead declares the maps in (41). Every consumed component is already committed and certificate-bound. Returned current, later recurrence and Duad seating retain that identity.

$$\text{Dimonon} = 2\,\text{Monon} \neq \text{Dion}, \qquad \text{Duon has the support of (30).} \tag{34}$$

The carried monon pivot μ_M and its state are distinct from omission of the mandatory temporal hinge. Dion endpoint adjacency never licenses a direct temporal branch reversal. Duad names a seated state of the same Duon current; a generic two-cycle C_2 supplies neither the Duon grammar nor its witness. Standalone trion names the threefold core unit; $3 : j$ composes j such units with inherited construction provenance.

3.5 Curling-curl support forms

A compact specification $p{:}q{:}L$ proposes a retained support form $p{:}q$ with enclosure/window qualifier L. Admission uses (33). Declare the finite support model

$$\Pi(p{:}q) = q^p + q. \tag{35}$$

Its edge-shell derivation is in App. H.5. With symmetric branches and one hinge, (32) gives $(\Pi, RD_B^{(1)}) = (4, 9), (10, 21), (30, 61), (68, 137)$ for 1:2, 3:2, 3:3, 3:4, respectively. Trace, kernel choice, hinge slide, jitter, capacity and admission remain explicit.

Definition 8 (Support expression)**.** *A support expression is formal syntax*

$$\Sigma = (s_1, \ldots, s_m; L), \tag{36}$$

where each $s_i = p_i{:}q_i$ and L is an optional shared enclosure qualifier.

Definition 9 (Curling-curl support form)**.** *A curling-curl support form is the retained recursive curl/cycle form proposed by a support expression and admitted on a declared boundary/window after its construction trace is declared or inherited.*

A compact expression such as 1:2:6, 3:3:6, or 3:4:6 proposes a specification K_{prop}. Admission is a traced calculation on the declared scope:

$$K_{\text{prop}} \xrightarrow{HT(K;B)} K_{\text{adm}}(B). \tag{37}$$

Here $HT(K; B)$ is the construction trace. Closure, pressure, shedding and comparison values follow that trace; a specification alone supplies none.

The downstream value construction is given in (135), after its duration, pressure and response interfaces.

Examples of proposed specifications are

$$1{:}2{:}6 = \text{Dimonhexon} = \text{Dimonon curling-curl support form carried on } C_6,$$

and

$$3{:}3{:}6 = \text{Tritriohexon}, \qquad 3{:}4{:}6 = \text{Tetratriohexon},$$

on C_6. SADAR couples their retained beats after admission.

Provenance. (40), (31), (92).

3.6 Scoped field organization

A field organizes admitted recurrent wave identities at a declared scope:

$$F_{B,\lambda}|_k = (B, \lambda, I_B(k), \{W^{\mathrm{id}}_{B,i}\}_{i \in I_B(k)}). \tag{38}$$

Here (B, λ) is field identity and $I_B(k)$ is present membership. Member state remains in S^W, not in the identity. Alias resolution maps names to this record and then to unique wave identities; two aliases of one wave do not supply two cadences. Existing certificates validate membership. That validation adds no closure, recurrence or wavehood.

$$C^{\mathrm{ret}} = \text{retained closure lineage}, \qquad \mathrm{Resolve}(F_{B,\lambda}) = \{W^{\mathrm{id}}_{B,i}\}_{i \in I_B}. \tag{39}$$

Memorum. The organization exposes members' retained predecessor closures, present boundaries and successor horizons. Finer wave organizations can be retained inside it. Their membership and evidence persist when a next enclosing scope is proposed.

3.6.1 Region

Definition 10 (Region)**.** *A region is a declared retained part of a field. Regions carry current, closure, pressure, and returned-current orientation.*

3.6.2 Boundary

Definition 11 (Boundary)**.** *A boundary is the retained interface of a declared region or field. Boundary accounting records current and closure across the boundary.*

3.6.3 Tesseract-admissible field boundary

For a field, specialize (11) to $\partial_T F$ and $B = (\partial_T F, \omega)$.

3.6.4 Polychiral shell dressing

Definition 12 (Polychiral shell dressing)**.** *A polychiral shell dressing is field outer dressing. It carries Capital-Alpha/Omega return orientation, polarization imbalance, return duration, pressure, and SADAR coupling.*

Tritrio::Tetratrio::hexon. App. B.4.2. Dimonhexon. §3.5.

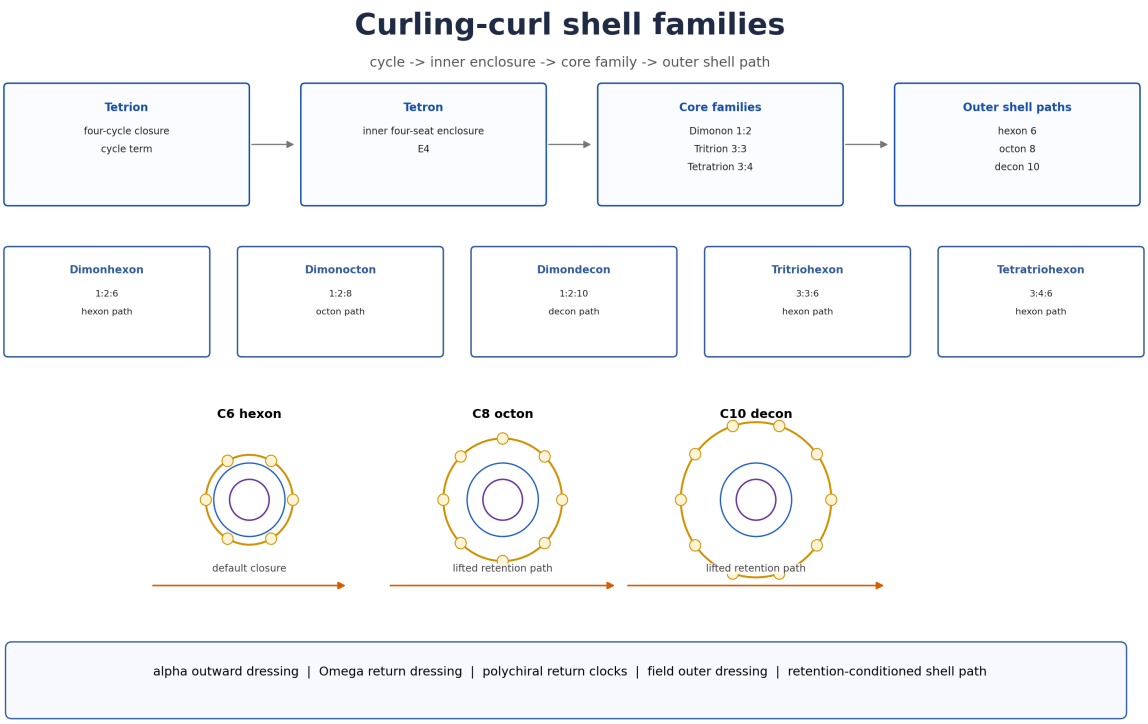


Figure 3: Curling-curl shell families.

Support-enclosure integrity. A tetron is E_4: the default inner enclosure with four seats. Hexon, octon, and decon name outer shell paths. App. B.3. A boundary can cut through a retained enclosure for accounting. That partial boundary is a sample of the enclosure structure. The complete enclosure identity follows the retained even-address topology.

3.7 Folded outer closure

A next-scope boundary enclosing retained wave organization $\mathcal{F}$ proposes an outer return. Conditional on its successful trace and certificate, $\mathrm{monon}_{\mathcal{F}}$ is the enclosing instance of (14), carrying the internal curls, shells, cycle presentations, duration, pressure and SADAR. This certified outer object is the premise of the compression (40); the full fold and exposure witnesses are (41) and (42).

3.8 Cycle-valued field compression

Definition 13 (Cycle-valued field compression)**.** *Let $f = \partial_T \mathcal{F}$ be a tesseract-admissible boundary and let $B_f = (f, \omega_f)$. For an internal support expression Σ, the certified cycle-valued compression, conditional on an actual outer return witness, is*

$$C_f[\Sigma]_{A\leftrightarrow\Omega} := \mathrm{cycle}_{\partial_T \mathcal{F}}\,[x \succ x = x;\ \mathrm{int} = \Sigma,\ \omega_f]_{A\leftrightarrow\Omega}\,. \tag{40}$$

The enclosing closure binds one next-scope closure-phase cycle; the boundary alone supplies a proposal. The internal support expression remains carried as cycle provenance.

Fold and exposure.

$$\{C^{\mathrm{ret}}_{B_\ell,i}\}_i \xrightarrow{\mathrm{Fold}^{\uparrow}} K^{\mathrm{prop}}_{B_{\ell+1}} \xrightarrow{\text{OuterClose+OuterCert}} C^{\mathrm{ret}}_{B_{\ell+1}}. \tag{41}$$

Contact, adjacency, superposition and mixing can propose an outer structure. The proposal binds the exact inner identities, their certificates and the map from each inner scope to the proposed outer scope. The outer certificate commits that construction witness; attaching an unrelated higher-scale loop afterward does not establish the fold. Each consumed inner input precedes its consumption. The certified outer closure contributes one complete closure-phase cycle. Its distinct compatible outer reclosures are required for an outer wave and then field organization.

$$\mathrm{Expose}^{\downarrow}(C^{\mathrm{ret}}_{B_{\ell+1}}) = \{C^{\mathrm{ret}}_{B_\ell,i}, \text{lineage}, \text{orientation}, \text{history}, \text{certificates}\}_i. \tag{42}$$

For an outer wave the same exposure additionally returns each inner wave, cadence, self-separation, current and memory. Outer shedding preserves surviving certified inner closures and waves. Inner beats cannot be reused as new outer returns. Fusion applies this closure construction to a folded complex; its pressure-resolved routing model is derived in Section 5.9.

Specialize (31) to the certified single-support object:

$$RD_{A\leftrightarrow\Omega}(C_f[p{:}q]; B_f) = \Pi^-_f(p{:}q) + H_{\mu,f}(p{:}q) + \Pi^+_f(p{:}q). \tag{43}$$

The subsequent $RD_{A\leftrightarrow\Omega}, \rho^D_\omega, P^D, Q^D$ accessors consume $C_f[\Sigma]_{A\leftrightarrow\Omega}$; none replaces that certified object.

3.9 Alpha–Omega hinge kernel

Definition 14 (Alpha–Omega hinge kernel)**.** *The explicit hinge presentations are*

$$A \leftrightarrow_\mu \Omega := \textit{Alpha/Omega branch order through hinge } \mu,$$
$$\Omega \leftrightarrow_\mu A := \textit{Omega/Alpha branch order through hinge } \mu.$$

For a retained cycle C, define the compact branch/hinge/branch operators

$$\mathsf{AO}_\mu(C) := \mathrm{A} \leftrightarrow_\mu \Omega(C) = \langle \Pi_\alpha(C), H_\mu(C), \Pi_\Omega(C)\rangle\,. \tag{44}$$

$$\mathsf{OA}_\mu(C) := \Omega \leftrightarrow_\mu \mathrm{A}(C) = \langle \Pi_\Omega(C), H_\mu(C), \Pi_\alpha(C)\rangle\,. \tag{45}$$

The hinge μ supports handed return. The chirality tag records the branch-order orientation:

$$\chi_\mu(C) = +1 \Longleftrightarrow C : A \leftrightarrow_\mu \Omega, \qquad \chi_\mu(C) = -1 \Longleftrightarrow C : \Omega \leftrightarrow_\mu A. \tag{46}$$

The hinge flip exchanges the branch order:

$$\mathrm{flip}_\mu(A \leftrightarrow_\mu \Omega) = \Omega \leftrightarrow_\mu A, \qquad \mathrm{flip}_\mu(\Omega \leftrightarrow_\mu A) = A \leftrightarrow_\mu \Omega. \tag{47}$$

The compact qualifier $A{\leftrightarrow}\Omega$ marks the Alpha–Omega hinge presentation in subscripts, as in $C_f[\Sigma]_{A\leftrightarrow\Omega}$. The forms $A\Omega$ and ΩA are implicit-hinge shorthand only after the explicit hinge presentation has been declared. The Ω-branch is closure-subservient:

$$\Pi_\Omega(C) = \mathrm{Close}_B(\Pi_\alpha(C), H_\mu(C)). \tag{48}$$

With branch support cardinalities, (31) *specializes to*

$$RD_{A\leftrightarrow\Omega}(C; B) = |\Pi_\alpha(C)| + H_\mu(C) + |\Pi_\Omega(C)|. \tag{49}$$

Equation (32) then gives the single-hinge specialization

$$H_\mu = 1, \qquad \Pi_\Omega = \Pi_\alpha, \qquad RD^{(1)}_{A\leftrightarrow\Omega} = 2|\Pi_\alpha| + 1. \tag{50}$$

When the hinge recurrence is explicitly attributed to free-seat provenance, write

$$H^{[s]}_\mu(C;B), \qquad s(C) = \text{declared free-seat provenance of } C. \tag{51}$$

For a full-seated single-hinge cycle, $H_\mu(C;B) = 1$; the notation $H^{\mathrm{cyc}}_\mu = 1$ attributes the hinge to that cycle. Free-seat provenance requires its separate declaration.

Tetratrion cycle accessor. For the certified $C_f[3{:}4]_{A\leftrightarrow\Omega}$, the single-hinge symmetric model in App. H.5 gives $\Pi_\alpha = 68$ and $RD^{(1)}_{A\leftrightarrow\Omega} = 137$. The cycle object remains distinct from this scalar.

4 Wave state and relational temporal structure

4.1 Complete state before comparison

Wave identity is fixed by (24). Its mutable state retains all coordinates needed by the declared realization:

$$\begin{aligned} S^W_{B,i,k} = (&W^{\mathrm{id}}_{B,i}, K^{\mathrm{adm}}_i, \sigma^C, J, p, \partial B, \Phi, \mathrm{Cad}, \varepsilon, \\ &\mathrm{memory}, \mathrm{capacity}, \mathrm{context})_k. \end{aligned} \tag{52}$$

The closure state σ^C may be forming, open, pending, closed, fusing, reclosed or shedding. Retained context has its own coordinate; it is not identical with closure memory or pressure. The companion coupling state is

$$S^\times_{I,B,k} = (E, \psi^{\mathrm{bias}}, \Lambda, \Delta\Lambda, \mathrm{RCD}, \rho^D, \mathrm{provenance})_k. \tag{53}$$

Its fields hold committed comparison records, with absent coordinates typed as absent until their preconditions have been established.

Declared update and trace. Declare an update law on its admissible input domain:

$$S^W_{k+1} = \mathcal{U}_B(S^W_{\le k}, S^\times_{\le k}, \mathrm{input}_{\le k}), \qquad \Gamma^W_{B,i}[k_0{:}k_1] = (S^W_{B,i,k})_{k_0\le k\le k_1}. \tag{54}$$

In the native event realization, state and input cuts are integers. Initialization at a completed return retains that return's vertex coordinate; any rebasing uses a declared coordinate map. This interface permits different declared realizations. Pair events have separate immutable records:

$$\begin{aligned} e^\times_s = (&\mathrm{id}_s, W^{\mathrm{id}}_i, W^{\mathrm{id}}_j, B, \omega, \mathrm{direction}, \mathrm{contact}, \mathrm{reflection}, \\ &k_s, \mathrm{digest}\,\Gamma^W_{i,\le k_s}, \mathrm{digest}\,\Gamma^W_{j,\le k_s}), \qquad \Gamma^\times = (e^\times_s)_s. \end{aligned} \tag{55}$$

Comparison requires both recurrent lineages, their common committed window and repeated distinct bound contacts; one contact supplies only a stage.

4.2 Directional RCD and duration

The scoped address binder is $\mathbf{a} \vdash (\mathcal{F}, R, \partial R, \omega)$. The field resolves to its wave members before a channel is evaluated.

Constructor. Couple the support accessor (31) before projecting duration:

$$\mathrm{RCD}_{ij}(B) = RD_B(C_{ij}) \bowtie_B R_{ij}^{\mathrm{asym}}(B), \tag{56}$$

$$\mathrm{dur}_{\mathcal{P}} : \mathrm{RCD}_{ij} \longrightarrow \mathcal{D}_{ij}, \qquad \rho_{ij}^{D}(B) = \mathrm{dur}_{\mathcal{P}}(\mathrm{RCD}_{ij}(B)). \tag{57}$$

Here $\mathcal{D}_{ij}$ is a declared exact ordered native count domain. The constructor retains both lineages, direction, repeated contact identities, scope and reflection witness. An additive-count specialization has domain $(RD, r_1, \ldots, r_m)$ with $r_s \geq 0$ and sets

$$\mathrm{dur}_{+}(RD \bowtie (r_s)_s) = RD + \sum_s r_s. \tag{58}$$

This is a declared duration specialization. Neutral reflection $r_s = 0$ specializes it to $\rho^D = RD$; the r_s depend on direction, so reversal can change duration.

For returned channel e, the retained duration coordinate is

$$D_e(\partial F, \omega) = \mathrm{dur}(\mathrm{RCD}_e), \tag{59}$$

with the specific duration interface declared on that channel.

Window participation. A window projection requires a compatible count in the same ordered domain:

$$\mathrm{Compat}_{\mathcal{D}}(\rho_{ij}^{D}, |\omega|_{ij}^{\mathcal{D}}) \quad \Longrightarrow \quad \rho_{\omega,ij}^{D} = \min_{\mathcal{D}}(\rho_{ij}^{D}, |\omega|_{ij}^{\mathcal{D}}). \tag{60}$$

Clipping changes only scalar participation. RD and RCD stay intact; support-enclosure retention TTL has its own interface in (113). Native relational time is constructed in (64).

4.3 Comparison from committed recurrence

The phase accessor is (22). A native comparison window has two strictly ordered integer endpoints $u < v$; contact commitments are integer cuts in $[u, v]$, and reclosures are counted in $(u, v]$ as in (23). For distinct recurrent lineages with repeated bound contacts in that window, the exact cadence ratio is

$$\mathrm{biz}_{i|j}^{TF}(\omega) = \frac{\Delta\,\mathrm{bip}_i(\omega)}{\Delta\,\mathrm{bip}_j(\omega)}, \qquad \Delta\,\mathrm{bip}_j(\omega) > 0. \tag{61}$$

Here $\Delta\mathrm{bip}_i(\omega) = \mathrm{bip}_i(\omega) = \mathrm{bip}_i^{\leq}(v) - \mathrm{bip}_i^{\leq}(u)$ uses the window accessor. A single monon pair supports structural comparison, but lacks the two cadences required here. A declared comparison protocol $\mathcal{P}$ fixes the exact ordered domain, orientation and exchange law:

$$\mathrm{Compare}_{\mathcal{P},i|j}^{\Phi}(\Phi_i, \Phi_j, \mathrm{Cad}_i, \mathrm{Cad}_j, \Gamma_{ij}^{\times}) = (\psi_{i|j}^{\mathrm{bias}}, \Lambda_{i|j}, \Delta\Lambda_{i|j}). \tag{62}$$

The native comparison is unwrapped. One declared count realization is

$$\psi_{i|j} = b_i - b_j, \quad \Lambda_{i|j} = N_i - N_j + b_i - b_j, \quad \Delta\Lambda_{i|j} = \Lambda_{i|j}(k) - \Lambda_{i|j}(k^{-}), \tag{63}$$

Here N_i, N_j count committed reclosures and $(b_i, b_j) \in \mathbb{Q}^2$ are separately declared ordered oriented offsets. A whole-closure slip changes Λ by one even if a wrapped display repeats. Another declared convention is the cadence-qualified $qX_i - pX_j$, whose reversal retains its coefficient normalization.

$$\begin{aligned}\mathrm{RT}_{I,B}(\omega) &= \Pi_{\mathrm{rel}}^{\mathrm{native}}(\{\Gamma_{B,i}^{W}\}_{i\in I}, \Gamma_{I,B,\omega}^{\times}; \mathcal{P}) \\ &= \{(\mathrm{biz}_{i|j}^{TF}, \psi_{i|j}, \Lambda_{i|j}, \Delta\Lambda_{i|j}, \mathrm{RCD}_{ij}, \rho_{ij}^{D}, \mathrm{provenance}_{ij})\}_{(i,j)\in E}.\end{aligned} \tag{64}$$

With positive counts in both directions, cadence ratios reverse reciprocally. The count comparator reverses bias and lag by sign; the directional RCD is reconstructed. Three or more waves form an edge-labelled comparison structure. No global scalar clock is required. Equal temporal coordinates may coexist with unequal pressure, closure condition or retained context in (52).

Falsification. Aliased wave identities, an uncoupled pair, one contact, zero denominator, unbound prefix, absent duration interface or incompatible clipping domain invalidate the stated coordinate. Aggregation that discards direction or ancestry loses part of the relation even when its current sums to zero.

Provenance. (21), (24), (54), (55), (56).

4.3.1 Tick/click

Specialize (54) with complete state K_t, emission E, received returned Duon Y_t, bias β_t and committed lead–lag history:

$$X_t = E(K_t), \qquad K_{t+1} = U(K_t, Y_t, \beta_t, \Delta_{\mathrm{lead/lag},\le t}), \qquad X_{t+1} = E(K_{t+1}).$$

A tick emits; a click receives the bound return before the next tick.

4.3.2 Markovian and retained-history continuity

A Markovian continuity continues by present cycle state. A retained-history continuity carries prior return, lead–lag burden, jitter, shell residue, duration, or SADAR channel memory. Both forms carry wave continuity when distinction-information persists through recurring duon cycle form.

4.4 Downstream continuation model

Setup. On already admitted support, choose nonnegative rational weights. The atomic-traversal kernel (66) records H1 base changes. Fibre-only additional dwell has zero base traversals and uses the separate $0_H \to 0_D$ continuation on an already entered hinge; it is not called an atomic H1 event. With the typed continuation support split as

$$\mathrm{Adm}_B = \mathrm{Adm}_B^{\mathrm{H1}} \sqcup \mathrm{Adm}_B^{\mathrm{fib}}, \tag{65}$$

the first part contains atomic base traversals and the second contains zero-base-traversal hinge-fibre continuations. Define the atomic kernel by

$$P_B(e', \sigma' \mid e, \sigma) = \begin{cases} w_B(e', \sigma')/Z_B, & (e', \sigma') \in \mathrm{Adm}_B^{\mathrm{H1}},\ Z_B > 0, \\ 0, & (e', \sigma') \notin \mathrm{Adm}_B^{\mathrm{H1}}, \end{cases} \qquad Z_B = \sum_{a \in \mathrm{Adm}_B^{\mathrm{H1}}} w_B(a). \tag{66}$$

For $Z_B = 0$, report empty positive-weight support; distinguish it from an empty structural admissible set. Neither case is normalized. Equal positive weights give $1/N_B$; unequal weights give the anisotropic model. Weights may use committed lag, orientation and reflection coordinates. Both choices preserve (9). Direct branch bypass has zero support. The kernel neither supplies the first distinction nor certifies closure from a phase threshold.

4.4.1 Hinge saturation

Setup. The additional-dwell continuation family and its unweighted/weighted empty conditions are

$$\mathrm{Adm}_B^0(e,\sigma) = \{(e,0_D) \in \mathrm{Adm}_B^{\mathrm{fib}}(e,\sigma)\}, \tag{67}$$

$$\mathrm{Adm}_B^0(e,\sigma) = \varnothing, \tag{68}$$

$$\sum_{(e,0_D)\in \mathrm{Adm}_B^{\mathrm{fib}}(e,\sigma)} w_B(e,0_D) = 0. \tag{69}$$

Derivation. After the mandatory hinge, any remaining admitted continuation then has

$$\sigma' \in \{-1,+1\}. \tag{70}$$

If that support is empty, the declared model reports pending, reseated, shedding or lifted state. Jitter records unresolved lawful continuation before seating. The ternary fibre has three roles; the branch-only image in (70) has at most two, regardless of its number of channel rows. For a declared binary role distribution $(z, 1-z)$, its conditional choice entropy is $-z\log_3 z - (1-z)\log_3(1-z) \le \log_3 2$: differentiation gives a unique maximum at $z = 1/2$, with the endpoints defined by continuity. This scoped entropy differs from (10) and general channel capacity.

Literature note. Hypercube support: Weisstein [10]; adjacency walks: MIT OCW [5], Duncan [1]; random walks: Norris [6]. Dyck/Catalan context: Stanley [9], Flajolet–Sedgewick [2].

5 Field and SADAR

The finite flux and response constructions retain their declared channel histories. For a committed value trace of (24), the harmonic representation begins at (139).

5.1 Typed finite flux

Setup. AFC declares normalized memory $p_i^{\mathrm{mem}} \ge 0$, $\sum_i p_i^{\mathrm{mem}} = 1$, and a stochastic kernel P. AΩD separately resolves pressure-bearing values of admitted returned channels. Keep their types explicit:

$$\phi_{ij}^{\mathrm{AFC}} = p_i^{\mathrm{mem}} P_{ij} - p_j^{\mathrm{mem}} P_{ji}, \qquad \phi_{ij}^{\mathrm{AOD}}(B) = p_{ij}^{D}(B) A_{ij}(B). \tag{71}$$

Each normalized product lies in $[0,1]$, hence $|\phi_{ij}^{\mathrm{AFC}}| \le 1$. Pressure values such as 7 are AΩD channel values in their declared count domain. Orientation reversal of a fixed antisymmetric flux changes its sign. A separately constructed reverse channel retains its own directional RCD and need not have the same magnitude.

Joint invariance. For a relabelling g, $p_{gi}^{\mathrm{mem}} = p_i^{\mathrm{mem}}$ and $P_{gi,gj} = P_{ij}$ imply $\phi_{gi,gj}^{\mathrm{AFC}} = \phi_{ij}^{\mathrm{AFC}}$ by substitution. Kernel symmetry alone supplies no such result: a two-state swap kernel with memory $(3/4, 1/4)$ has flux $1/2$, whereas equal memory gives zero.

Region and boundary. Let V be the finite vertex set, $R \subseteq V$, and $\partial R = \{(i,j) : i \in R, j \notin R\}$ its oriented cut.

$$\operatorname{div}(i;B) = \sum_{j \in V} \phi_{ij}(B). \tag{72}$$

5.1.1 Finite Stokes

Derivation. For either one declared antisymmetric flux type, split the double sum:

$$\begin{aligned} \sum_{i \in R} \operatorname{div}(i;B) &= \sum_{i,j \in R} \phi_{ij}(B) + \sum_{i \in R, j \notin R} \phi_{ij}(B) \\ &= \sum_{i \in R, j \notin R} \phi_{ij}(B). \end{aligned} \tag{73}$$

The internal sum is zero because each ordered pair cancels its reverse and diagonal entries are zero. Cancellation removes a summed value, not the two contributing lineage records. The same finite argument holds componentwise.

$$\phi^{\mathrm{AOD}}_{ij,A\leftrightarrow\Omega}(B) = p^{D}_{ij,A\leftrightarrow\Omega}(B) A_{ij,A\leftrightarrow\Omega}(B). \tag{74}$$

In the remaining pressure formulas ϕ abbreviates ϕ^{AOD}.

Falsification. Wrong antisymmetry or an omitted boundary channel breaks the finite identity.

Provenance. (56), (78).

Literature note. AFC Choices (memory, stochastic kernel, flux) and Stokes corollary, finite internal-pair cancellation: Poyau [8], Choices 1–5, Corollary 1 and equation (1), source revision `5a25e8bd5bbc`.

5.1.2 B-scoped flux contents

Resolve the AΩD channel operands of (71) on $B = (\partial F, \omega)$.

For a declared path-history family $\mathcal{H}_{ij}(B)$, the path-history pressure is

$$p^{D}_{ij}(B) = \sum_{h \in \mathcal{H}_{ij}(B)} C_{3,h} \rho^{D}_{\omega,h}(B). \tag{75}$$

A compressed channel form is licensed only after the declared compression

$$\mathsf{Comp}_{ij} : \{(C_{3,h}, RD_h, R^{\mathrm{asym}}_h, A_h)\}_{h \in \mathcal{H}_{ij}(B)} \mapsto (C_{3,ij}, \rho^{D}_{ij}, A_{ij}). \tag{76}$$

Under that declared compression,

$$p^{D}_{ij}(B) = C_{3,ij} \rho^{D}_{\omega,ij}(B) = C_{3,ij} \min\big(\rho^{D}_{ij}(B), |\omega|_{ij}\big). \tag{77}$$

The bare-duration form is the neutral-reflection reduction:

$$R^{\mathrm{asym}}_{ij}(B) = \mathrm{id} \quad \Rightarrow \quad \rho^{D}_{ij}(B) = D_{ij}(B).$$

The direction term is

$$A_{ij}(B) = \Delta\chi^{\mathrm{lag}}_{ij}(B) i_{ij}(B). \tag{78}$$

The AΩD branch of (71) gives

$$\phi_{ij}(B) = p^{D}_{ij}(B) A_{ij}(B). \tag{79}$$

Equivalently,

$$\phi_{ij}(B) = C_{3,ij} \rho^{D}_{\omega,ij}(B) \Delta\chi^{\mathrm{lag}}_{ij}(B) i_{ij}(B). \tag{80}$$

Magnitude and direction. In (79), p^D is magnitude and A is direction. Scaling every p^D by λ at fixed orientation and channel family scales both ϕ and its attention sum $\mathbf{A}_F$ by λ.

For a fixed antisymmetric flux on the oriented pair (i, j):

$$\phi_{ji}(B) = -\phi_{ij}(B), \qquad \phi_{ii}(B) = 0. \tag{81}$$

When $i_{ij}(B)$ has components, the antisymmetry and Stokes identity are evaluated componentwise.

The divergence is the sum defined in (72).

Specializing (73) to this flux gives

$$\sum_{i \in R} \mathrm{div}(i; B) = \sum_{\substack{i \in R \\ j \notin R}} \phi_{ij}(B). \tag{82}$$

SADAR as B-scoped AΩD flux		
Scope	$\to$	$B = (\partial F, \omega)$
RCD	$\to$	$\mathrm{RCD}_{ij}(B) = RD_B(C_{ij}) \bowtie_B R^{\mathrm{asym}}_{ij}(B)$
Scalar duration	$\to$	$\rho^D_{ij}(B) = \mathrm{dur}(\mathrm{RCD}_{ij}(B))$
Window clip	$\to$	$\rho^D_{\omega,ij}(B) = \min(\rho^D_{ij}(B), \lvert\omega\rvert_{ij})$
Magnitude	$\to$	$p^D_{ij}(B) = C_{3,ij}\rho^D_{\omega,ij}(B)$
Direction	$\to$	$A_{ij}(B) = \Delta\chi^{\mathrm{lag}}_{ij}(B) i_{ij}(B)$
Flux	$\to$	$\phi_{ij}(B) = p^D_{ij}(B) A_{ij}(B)$
Attention	$\to$	$\mathbf{A}_F(B) = \sum_{(i,j)\in F(B)} \phi_{ij}(B)$
Motion	$\to$	(S^W_k, S^W_{k+1}) under $\mathcal{U}_B$

Figure 4: SADAR as B-scoped AΩD flux.

Provenance. (73); (74); (79); (82).

5.1.3 Slice flux

Definition 15 (Duon slice flux)**.** *For a declared cut-frame t, let*

$$B|_t = (\partial F, \{t\}).$$

The duon-current slice flux is the one cut-frame crossing

$$\Phi^D_{\partial R}(t; B|_t) = \sum_{e \in \partial R(t)} \phi_e(B|_t). \tag{83}$$

When the scope is fixed, write $\Phi^D_{\partial R}(t)$. For raw duon-current content, $\phi_e(B|_t) = \epsilon_e C_{3,e}$.

5.1.4 Window current

Definition 16 (Window duon current)**.** *The window current is the retained many-frame continuation*

$$J^D_{\partial R}(\omega; B) = \sum_{t \in \omega} \Phi^D_{\partial R}(t; B|_t). \tag{84}$$

When the scope is fixed, write $J^D_{\partial R}(\omega)$.

The raw capacity crossing count is

$$N^D_{\partial R}(\omega; B) = \sum_{t\in\omega} \sum_{e\in\partial R(t)} C_{3,e}. \tag{85}$$

The magnitude current is

$$J^D_{\mathrm{mag}}(\omega; B) = \sum_{t\in\omega} \sum_{e\in\partial R(t)} p^D_e(B|_t). \tag{86}$$

When the declared boundary/window partitions current status, require

$$J^D_{\mathrm{mag}} = J^D_{\mathrm{in}} + J^D_{\mathrm{out}} + J^D_{\mathrm{pending}}. \tag{87}$$

5.1.5 Moving boundary

For a moving retained boundary, the window may record crossing, sweep, and shedding terms as

$$\Phi^D_{\partial R,\mathrm{mov}}(t) = \Phi^D_{\mathrm{cross}}(t) - \Phi^D_{\mathrm{sweep}}(t) + \Phi^D_{\mathrm{shedding}}(t). \tag{88}$$

Here crossing is duon current crossing the retained boundary, sweep is support swept by boundary motion, and shedding is disturbance that breaks from the retained boundary.

5.1.6 Regional boundary template

The same boundary/current form applies across familiar region choices. Declare a region, declare ∂R, and account for current crossing, returning, and remaining pending.

Duon-current boundary accounting

Internal retained	**Exoshedding**	**Reflected return**	**Pending return**
recurring core	crosses ∂R	returns through shell	remains over ω
scalar burden	shell crossing	next emission	duration overlap

$$\Phi^D_{\partial R}(t; B|_t) = \textstyle\sum_{e\in\partial R(t)} \phi_e(B|_t)$$
$$J^D_{\partial R}(\omega; B) = \textstyle\sum_{t\in\omega} \Phi^D_{\partial R}(t; B|_t)$$

Figure 5: Duon-current boundary accounting.

Provenance. (30), (83), (84), (88).

5.2 Field count

5.2.1 Scope

Fix a boundary/window scope

$$\mathcal{B} = (\partial F, \omega).$$

All quantities in this subsection are evaluated with respect to $\mathcal{B}$.

5.2.2 Retained field family

Let

$$\mathcal{F}_{\mathcal{B}} = \{F_1, \ldots, F_{n_F}\}$$

be the retained field family, and define

$$n_F(\mathcal{B}) := |\mathcal{F}_{\mathcal{B}}|. \tag{89}$$

Here n_F is retained field count, T is path/account duration or tick count, m is enclosure order in C_{2m}, and p is rotor or cycle completion period.

5.2.3 Returned-current channels

For $F_i, F_j \in \mathcal{F}_{\mathcal{B}}$, write

$$\mathrm{ADAR}_{i \leftarrow j}$$

for the returned-current channel from F_j into F_i. The diagonal terms are self-return terms; off-diagonal terms are cross-return terms. A retained field family may carry field-ring returned-current relations when E_{return} is declared on the fixed boundary/window.

5.2.4 Field-count term

For $n_F(\mathcal{B}) \geq 1$, the declared count-weighting model is

$$\mathcal{O}_{\partial F}(\mathcal{B}) = n_F(\mathcal{B}) \log n_F(\mathcal{B}), \tag{90}$$

with the logarithm base fixed by the declared calculation. A single scoped organization of an admitted recurrent monon-mode wave has $n_F = 1$. Alias names are resolved before counting; an isolated monon is not a field. Coupled-field counting begins at $n_F = 2$.

5.2.5 Disoma declaration

For a two-field declared scope, fix

$$B_{12} = (\partial F_{12}, \omega_{12}), \qquad \mathcal{F}_{B_{12}} = \{F_1, F_2\}.$$

Thus

$$n_F(B_{12}) = 2, \qquad \mathcal{O}_{\partial F}(B_{12}) = 2 \log 2.$$

Calculation split. Field-count scaling and shell/enclosure routing are connected by the declared boundary/window, but they are separate calculations:

$$B \to \mathcal{F}_B \to n_F(B) \to \mathrm{ADAR}_{i \leftarrow j} \to \mathcal{O}_{\partial F}(B) \to \mathrm{SADAR}.$$

Shell/enclosure routing fixes

$$\mathcal{C}_{2m} \to X_{\mathrm{shedding}} \to \mathrm{SheddicPath}(X_{\mathrm{shedding}}),$$
$$\mathrm{SheddicPath} \to \{\text{rebalancing},\ \text{exoshedding},\ \text{endoshedding},\ \text{redirected support routing}\}.$$

Scope $\mathcal{B} = (\partial F, \omega)$ and resolved organizations $\mathcal{F}_{\mathcal{B}} = \{F_1, \ldots, F_{n_F}\}$

$n_F = |\mathcal{F}_{\mathcal{B}}|$, declared count term $n_F \log n_F$

A single recurrent monon-mode wave organization has $n_F = 1$.
A lone monon closure supplies one complete closure-phase cycle; field membership requires recurrence.

Figure 6: Fixed-scope field count and SADAR scope.

Provenance. (89), (90), (97).

5.3 Duonic pressure

5.3.1 Scalar burden

Definition 17 (Channel pressure)**.** *Channel pressure is* (77) *on the declared compression. Duration participation is* (60)*; seat retention has the distinct interface* (113)*.*

Definition 18 (Duonic pressure)**.** *The scalar pressure of a field is*

$$P_F^D(B) = \sum_{e \in F(B)} p_e^D(B). \tag{91}$$

It records the retained/clipped duon/trit burden carried on the declared boundary/window; pending overhang is recorded separately by Q^D*.*

5.3.2 Pressure weighting

When $P_F^D(B) > 0$, a normalized orientation $\theta_F^{(q)}$ may be weighted by scalar pressure to provide scalar factors later used in $\mathbf{A}_F(B)$ in §5.5.

5.3.3 Inertial burden

Duonic pressure resists change of attention orientation. In the declared inverse-pressure response model with $P_F^D(B) > 0$, a committed perturbing current supplies the next-transition orientation increment:

$$\Delta\theta_{F,k+1}^{(q)} = \frac{\Delta J_{\text{duon}}^{(q)}}{P_F^D(B)}.$$

Provenance. (111), (91).

5.4 SADAR value operator

Definition 19 (SADAR operator)**.** *Write* $\mathfrak{S}_B$ *for Symmetry Attention Duration Asymmetric Reflection (SADAR) on the declared boundary/window* B*. For a retained cycle* C*,*

$$\mathfrak{S}_B(C) = \phi_{C,A\leftrightarrow\Omega}(B) = p_{C,A\leftrightarrow\Omega}^D(B) A_{C,A\leftrightarrow\Omega}(B). \tag{92}$$

Expanded,

$$\mathfrak{S}_B(C) = C_{3,C}\rho^D_{\omega,C}(B)\,\Delta\chi^{\mathrm{lag}}_C(B)i_C(B). \tag{93}$$

For two committed SADAR-coupled recurrent lineages C_u, C_v,

$$\mathfrak{S}_B(C_u, C_v) = \phi^{\mathrm{beat}}_{uv,A\leftrightarrow\Omega}(B) = p^D_{uv,A\leftrightarrow\Omega}(B)A_{uv,A\leftrightarrow\Omega}(B). \tag{94}$$

If $\bar{e}$ is the reversed oriented boundary element of e, then

$$\mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e). \tag{95}$$

5.5 SADAR

5.5.1 ADAR channel

Definition 20 (ADAR)**.** *ADAR is one local returned-current channel:*

$$\mathrm{ADAR}_{ij}(B) = (A_{ij}(B), D_{ij}(B), R^{\mathrm{asym}}_{ij}(B)). \tag{96}$$

Its operands are attention orientation (78)*, returned duration* (59)*, and the channel's asymmetric reflection on B.*

5.5.2 SADAR balance

Definition 21 (SADAR)**.** *SADAR is Symmetric Attention–Duration–Asymmetric Reflection. It is the field's symmetry-chasing returned-current dynamic on a declared boundary/window. For $B = (\partial F, \omega)$,*

$$\mathrm{SADAR}_F(B) = \big(\mathrm{ADAR}_{ij}(B)\big)_{(i,j)\in F(B)}. \tag{97}$$

The family retains incidence, phase relation and pressure. Relative balance uses attention, duration, asymmetric reflection, phase bias, reseating and shedding.

5.5.3 SADAR attention vector

Definition 22 (SADAR attention vector)**.** *The SADAR attention vector is the boundary/window-specific codification of SADAR dynamics. It records pressure-weighted response input of returned duon current on the declared boundary/window:*

$$\mathbf{A}_F(B) = \sum_{(i,j)\in F(B)} \phi_{ij}(B). \tag{98}$$

Equivalently,

$$\mathbf{A}_F(B) = \sum_{(i,j)\in F(B)} C_{3,ij}\rho^D_{\omega,ij}(B)\Delta\chi^{\mathrm{lag}}_{ij}(B)i_{ij}(B). \tag{99}$$

The expanded sum substitutes (80); pressure and orientation retain their separate channel types.

5.5.4 Phase bias, push/pull, and motion

Phase bias is the signed lead–lag imbalance of returned-current channels. Push and pull are signed phase-biased components of the same SADAR attention vector.

Realized motion. The pressure-bearing attention vector is an input to the declared update. Motion is its realized state transition:

$$\mathrm{motion}_{F,k+1}(B) = (S^W_{F,k}, S^W_{F,k+1}), \qquad S^W_{F,k+1} = \mathcal{U}^{\mathrm{SADAR}}_B(S^W_{\le k}, S^{\times}_{\le k}, \mathbf{A}_{F,\le k}). \tag{100}$$

All pressure, lag and directional-duration inputs are committed through k; the response is at $k+1$. No coordinate is generated by the same response that consumes it. For a declared rational example with $p_k > 0$,

$$J_{k+1} = J_k + \mathbf{A}_k/p_k. \tag{101}$$

This realization leaves identity and closure memory fixed unless a separate admissible closure transition is witnessed. A zero scalar residual alone is not such a witness. The support triples AO and OA exchange branches; their oriented signed values obey $AO = -OA$, with the hinge count retained.

Statement: incident return shift. If $\mathbf{A}_F(B) = 0$ and an unmatched incident returned duon e contributes

$$\phi_e(B) = p^D_e(B)A_e(B),$$

then

$$\mathbf{A}_{F+e}(B) = \phi_e(B). \tag{102}$$

If a matching return $\bar{e}$ is declared and $\phi_{\bar{e}}(B) = -\phi_e(B)$, then

$$\mathbf{A}_{F+e+\bar{e}}(B) = 0. \tag{103}$$

Statement: short window and high incident current. If $|\omega|_h < \rho^D_h(B)$, then $\rho^D_{\omega,h}(B) = |\omega|_h$ and $Q^D_h(B) > 0$. For a short window,

$$p^D_h(B) = C_{3,h}|\omega|_h.$$

Many incident histories or large $C_{3,h}$ can still shift the SADAR attention vector when

$$\sum_h p^D_h(B)A_h(B) \neq 0.$$

The channel-level rate form is

$$\Delta\phi_{ij} = p^D_{ij}\Delta A_{ij}$$

when retained pressure is fixed across the compared restrictions. Pending overhang separates from clipped pressure:

$$Q^{D,\mathrm{dur}}_{ij}(B) = \max\bigl(0, \rho^D_{ij}(B) - |\omega|_{ij}\bigr), \tag{104}$$

$$Q^D_{ij}(B) = C_{3,ij}Q^{D,\mathrm{dur}}_{ij}(B). \tag{105}$$

A falling field may remain internally balanced relative to its own boundary/window while its SADAR attention vector is phase-biased relative to another declared boundary.

5.5.5 Self-return and cross-return

A single declared field carries self-return symmetry:

$$\mathrm{ADAR}_{i\leftarrow i} = \text{self-return channel.}$$

When another field is included in the same accounting scope, cross-return channels are added:

$$\mathrm{ADAR}_{i\leftarrow j} = \text{cross-return channel}, \qquad i \neq j.$$

Self-return and cross-return are channel forms inside the same SADAR balance. The field ring gate has the declared boundary gates, returned-current relations and value domain of §5.10.

5.5.6 Boundary-relative simultaneity

Boundary-relative simultaneity names the coexistence of component-field and enclosing-field SADAR balances. Component fields and enclosing fields may each carry their own SADAR attention vector because each vector is scoped by its declared boundary/window. For the enclosing two-field scope, write

$$\mathcal{B}_{12} = (\partial F_{12}, \omega_{12}).$$

Then

$$\mathbf{A}_{F_1}(B_1), \qquad \mathbf{A}_{F_2}(B_2), \qquad \mathbf{A}_{F_1\|F_2}(\mathcal{B}_{12}).$$

For three fields, write

$$\mathcal{B}_{123} = (\partial F_{123}, \omega_{123}),$$

and

$$\mathbf{A}_{F_1}(B_1), \quad \mathbf{A}_{F_2}(B_2), \quad \mathbf{A}_{F_3}(B_3), \quad \mathbf{A}_{F_{12}}(\mathcal{B}_{12}), \quad \mathbf{A}_{F_{123}}(\mathcal{B}_{123}).$$

Component scopes. $\mathbf{A}_{F_1}(B_1)$ and $\mathbf{A}_{F_2}(B_2)$ retain their own boundary/window and wave participants.

Enclosing scope. $\mathbf{A}_{F_1\|F_2}(\mathcal{B}_{12})$ is evaluated on its declared enclosing boundary after recurrent membership is resolved.

Committed attention and later motion.

$$\mathbf{A}_{F,k}(B) = \sum_{e\in F(B)} C_{3,e}\rho^{D}_{\omega,e}(B)\Delta\chi^{\mathrm{lag}}_{e,k}(B)i_{e,k}(B),$$

$$S^{W}_{F,k+1} = \mathcal{U}^{\mathrm{SADAR}}_{B}(S^{W}_{\leq k}, S^{\times}_{\leq k}, \mathbf{A}_{F,\leq k}).$$

The realized transition is motion. Unresolved excess follows a declared sheddic path. Component projections are taken from the already declared attention vector; they do not create a separate motion law.

Figure 7: SADAR boundary-relative balance. Component fields carry local SADAR attention vectors on their own declared boundaries. An enclosing boundary binds them as a super-field with its own SADAR attention vector. Phase bias shifts the boundary/window-specific SADAR balance; unresolved excess follows a sheddic path.

5.5.7 Sheddic route reference

Computed excess follows (108) and the declared route (109).

5.5.8 Component projection

Component projections select declared components of the already-fixed SADAR attention vector.

Provenance. (96), (97), (98), (56), (100), (102), (103), (109), App. B.

5.6 Cycle-shedding and temporal SADAR burden

5.6.1 Temporal cycle group

Definition 23 (Temporal cycle group)**.** *A temporal cycle group is a carried segment of duon current participating in wave continuity. App. F.*

5.6.2 Lead-lag burden

Definition 24 (Lead–lag disturbance span)**.** *The lead–lag disturbance span is the carried asymmetric return extent of a temporal cycle group on the declared cut lineage.*

5.6.3 Temporal SADAR burden

For a temporal cycle group W on a declared boundary/window, its temporal SADAR burden is

$$P_W^{\mathfrak{S}}(\partial F, \omega) = \sum_{e \in W} C_{3,e} \rho_{\omega,e}^{D}(\partial F, \omega) \left| \Delta_e^{\chi^{\mathrm{lag}}} \right|. \tag{106}$$

This burden is taken up by SADAR balance in §5.5.

5.6.4 Shedding

Definition 25 (Shedding)**.** *Shed is the verb. Shedding is the direction-neutral noun for the parent surplus process. Sheddic is the adjective for path, route, value, and audit descriptors. Endoshedding and exoshedding carry directionality.*

Shedding is the break of excess carried disturbance from local closure compatibility. If $C_W^{\mathrm{close}}(\partial F, \omega)$ is local closure compatibility, then

$$\Delta_{\mathrm{close}}(\partial F, \omega) = P_W^{\mathfrak{S}}(\partial F, \omega) - C_W^{\mathrm{close}}(\partial F, \omega). \tag{107}$$

$$X_{\mathrm{shedding}}(\partial F, \omega) = \max(0, \Delta_{\mathrm{close}}(\partial F, \omega)) \tag{108}$$

records sheddic excess. A negative closure residual is a deficit/open check, not sheddic surplus.

5.6.5 Sheddic path

Seat-compatible retention places the computed (108) on a declared path:

$$\mathrm{SheddicPath}(X_{\mathrm{shedding}}) \in \{E_2, \mathsf{C}_6, \mathsf{C}_8, \mathsf{C}_{10}, \ldots, \mathsf{C}_{2n}\}. \tag{109}$$

After a route partition is declared, the parent surplus may be placed as

$$X_{\mathrm{shedding}} = X_{\mathrm{endo}} + X_{\mathrm{exo}} + X_{\mathrm{redir}} + X_{\mathrm{open}}. \tag{110}$$

The terms are endoshedding, exoshedding, redirected support and unretained open loss, assigned only after declaring SheddicPath. With reclosure fraction $\lambda_{\mathrm{reclose}} \in [0,1]$, write

$$X_W^{\mathrm{reclose}} = \lambda_{\mathrm{reclose}} X_{\mathrm{shedding}}, \qquad X_W^{\mathrm{out}} = (1-\lambda_{\mathrm{reclose}}) X_{\mathrm{shedding}}.$$

Here X_W^{out} is the non-reclosed outward remainder. After route placement,

$$X_W^{\mathrm{out}} = X_{\mathrm{exo}} + X_{\mathrm{redir}} + X_{\mathrm{open}}.$$

A sheddic path may reclose, continue outward, or carry decay status according to seat-compatible retention and support-enclosure compatibility.

5.6.6 Yro odd-window route-status suffix

For an odd declared window length $L = 2k+1$, write

$$\mathrm{Yro}_L(K; B).$$

This is an odd-window unstable route-status suffix for a compact curling-curl specification K on the declared boundary/window B. It records route status between adjacent even support enclosures C_{2k} and C_{2k+2}. The support-enclosure ladder remains the even sequence

$$\mathsf{C}_6,\ \mathsf{C}_8,\ \mathsf{C}_{10}, \ldots, \mathsf{C}_{2n}.$$

The yro status may be driven by surplus, deficit, pending overhang, hinge slide, seat-compatible retention cutoff, terminal split, or another declared route instability. The construction trace and sheddic-route audit fix which condition applies on the declared scope.

Naming audit. A yro suffix is audited as route status on a declared boundary/window and is recorded separately from support-enclosure names.

5.6.7 Entropic duon current

Definition 26 (Entropic duon current)**.** *Entropic duon current is retained duon current carrying unresolved return burden. It carries retained ternary capacity through shell return, shedding, reclosure, and SADAR dynamics.*

For a declared field, boundary, and window, one local scalar form is

$$J_{\mathrm{ent}}^D(\partial F, \omega) = \sum_{e \in F(\partial F, \omega)} C_{3,e} \rho_{\omega,e}^D(\partial F, \omega) \left| \Delta_e^{\chi^{\mathrm{lag}}} \right|. \tag{111}$$

The sum specializes (106) to the field's retained current family. Coherent continuity can accumulate the unresolved lead–lag burden.

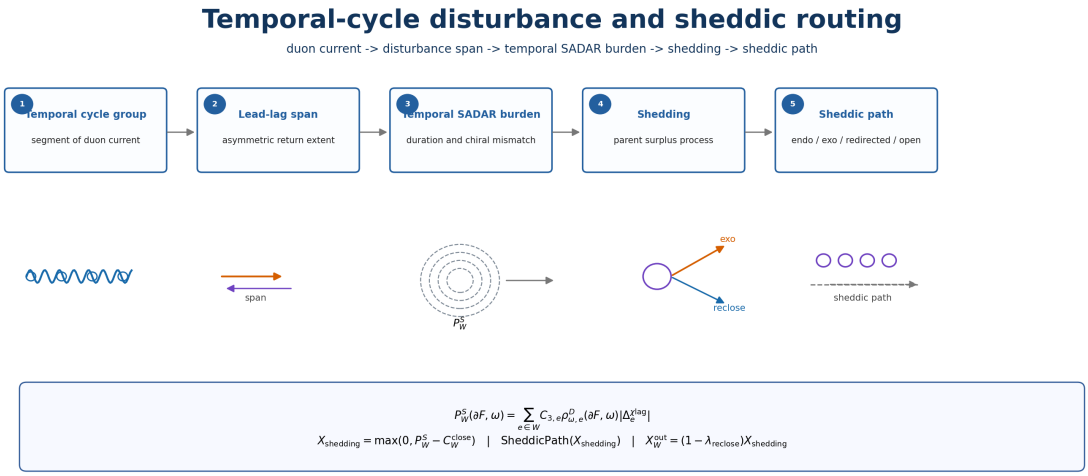


Figure 8: Temporal cycle lead–lag disturbance and shedding.

Provenance. (30), (59), (106), (108), (109), (98), (111), (97), (112), App. F, App. B.

5.7 Seat compatibility and shell/enclosure routing

5.7.1 Even sequence

The even sequence separates the current-side double form, the inner enclosure, and the outer enclosures.

5.7.2 E2 duon

$$E_2 = \text{duon}, \qquad E_2^0 = \text{open-seat duon}.$$

5.7.3 E2$^{\pm}$ duad

$$E_2^{\pm} = \text{duad}.$$

Duad is the seated state of E_2.

5.7.4 E4 tetron

$$E_4 = \text{tetron}.$$

Tetron is the default inner enclosure with four seats.

5.7.5 Outer enclosures

A support enclosure is an outer even enclosure available to shell/enclosure routing:

$$\mathsf{C}_{2m} = \text{support enclosure of order } 2m, \qquad m \geq 3.$$

5.7.6 Hexon

$$\mathsf{C}_6 = \text{hexon}.$$

Hexon is the first outer enclosure because the hinge triad supplies left/right threefold closure: $\mathsf{H}_3 + \mathsf{H}_3 = 6$.

5.7.7 Octon

$$\mathsf{C}_8 = \text{octon}.$$

5.7.8 Decon

$$\mathsf{C}_{10} = \text{decon}.$$

5.7.9 Higher even enclosures

Higher even enclosures continue the outer sheddic path sequence:

$$\mathsf{C}_{2n}, \qquad n > 5.$$

Outer enclosures are retained forms of sheddic paths. Hexon is the first outer enclosure; octon and decon are higher sheddic path forms; C_{2n} records the higher even sequence.

5.7.10 Filled shared enclosure

A filled shared enclosure has

$$m : m : 0 :: 2m, \qquad m \geq 3.$$

Examples:

$$3 : 3 : 0 :: 6 = \text{Tritrio}^3 :: \text{Tetratrio}^3 :: \text{hexon},$$
$$4 : 4 : 0 :: 8 = \text{Tritrio}^4 :: \text{Tetratrio}^4 :: \text{octon},$$
$$5 : 5 : 0 :: 10 = \text{Tritrio}^5 :: \text{Tetratrio}^5 :: \text{decon}.$$

5.7.11 Closure excess

For a support-enclosure declaration $\mathcal{C} = a : b : c :: L$ on $B = (\partial F, \omega)$, shell/enclosure routing carries the already-declared sheddic excess on the available route $\mathcal{C}$. The routed closure excess is

$$X_{\mathcal{C}}(B) = \text{route}_{\mathcal{C}}(X_{\text{shedding}}(B)) \,. \tag{112}$$

The value and placement of $X_{\mathcal{C}}$ determine local rebalancing, exoshedding, endoshedding, or redirected support routing.

5.7.12 Seat-compatible retention duration

For a support-enclosure declaration $\mathcal{C} = a : b : c :: L$, write

$$\mathrm{TTL}_{\mathcal{C}}(\partial F, \omega) = \text{retained tick duration available for seat-compatible SADAR rebalancing.} \tag{113}$$

This is distinct from $\rho^{D}_{\omega,ij}(B)$, the window-clipped duration participation clip for a channel. Retention on the declared window is $\mathrm{TTL}_{\mathcal{C}}(\partial F, \omega) \geq |\omega|$. If $\mathrm{TTL}_{\mathcal{C}}(\partial F, \omega) < |\omega|$, the sheddic path continues through re-entry, reclosure, or decay status before the window closes.

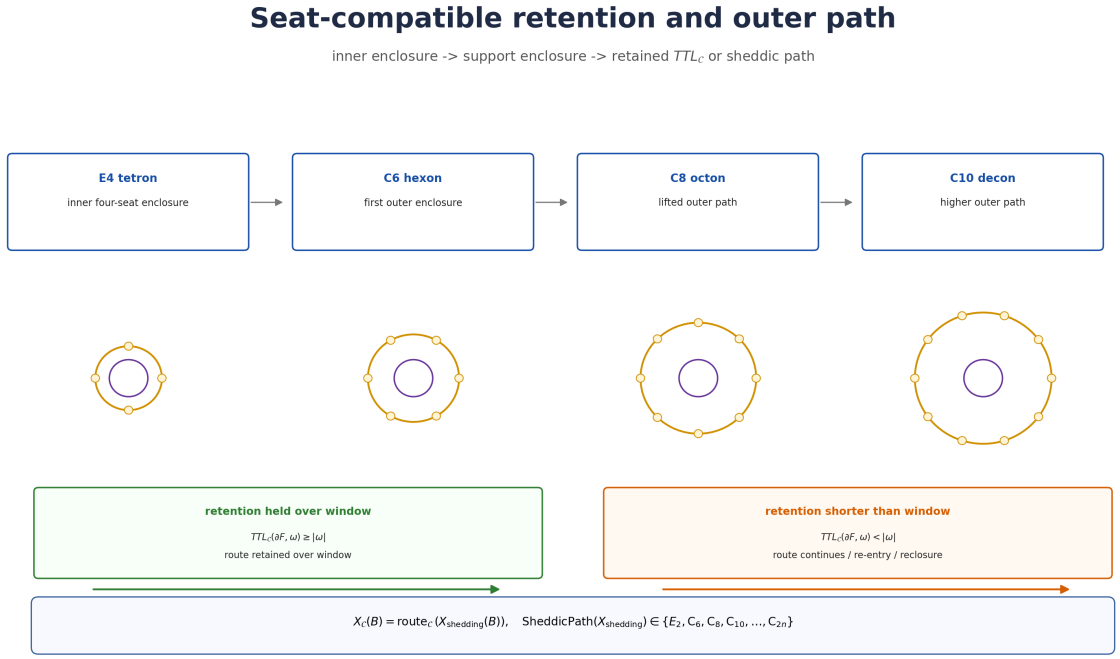


Figure 9: Seat compatibility and support-retention duration for sheddic paths.

5.7.13 Coord-0 grouping

At coordinate 0, write coord-0 shell-stack declarations as

$$0 \mid - :: Z : A : 0 : L\langle a : b : c :: L\rangle.$$

For filled support-enclosure grouping,

$$0 \mid - :: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle.$$

Define the cumulative group boundary

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0.$$

Then $\Gamma_{m-1} < Z \leq \Gamma_m$ selects the coord-0 support-enclosure group.

Provenance. (97), (112), (113), App. B.

5.8 Shell paths

5.8.1 Shared shell-path declaration

The shared shell-path declaration may be written

$$0 \mid - :: Z : A : 0 : 6\langle 1 : 1 : 0 :: 6\rangle.$$

Named:

$$0 \mid - :: Z : A : 0 : \text{hexon}\langle \text{Tritrio} :: \text{Tetratrio} :: \text{hexon}\rangle.$$

For the filled hexon field, write

$$0 \mid - :: Z : A : 0 : 6\langle 3 : 3 : 0 :: 6\rangle.$$

Named:

$$0 \mid - :: Z : A : 0 : \text{hexon}\langle \text{Tritrio}^3 :: \text{Tetratrio}^3 :: \text{hexon}\rangle.$$

5.8.2 Nested shell-stack pattern

The same curling-curl pattern iterates upward through shared enclosures and filled enclosures:

$$\text{Tritrio}^3 :: \text{Tetratrio}^3 :: \text{hexon} \to \text{Tritrio}^4 :: \text{Tetratrio}^4 :: \text{octon} \to \text{Tritrio}^5 :: \text{Tetratrio}^5 :: \text{decon}.$$

5.8.3 Shell-path closure references

Shell paths retain channel pressure (77) and flux (79). Their closure controls are in App. H and App. I.

5.9 Fusion closure and chain reclosure

5.9.1 Fusion closure

Fusion is branch/hinge/branch closure of admitted curling-curl support forms. For admitted forms F and G, define

$$\text{Fuse}_B(F, G) := \text{Close}_B\left(\widehat{F}\ \mu\ \widehat{G}\right), \tag{114}$$

where $\widehat{F} = \text{Compress}(F)$ and $\widehat{G} = \text{Compress}(G)$. Closed forms fuse; exposed fragments do not. The inherited fusion operator consumes closed or reclosed admitted inputs. Each input compression must have its complete inner construction and certificate bound before consumption. The outer certificate binds both compressed inputs; an incomplete compression or an open proposal is outside this operator's domain.

The branch/hinge/branch fusion duration is

$$RD_{\text{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu,\widehat{F},\widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{115}$$

For a closure rung,

$$C^{\text{ret}}_{n+1} = \text{Close}_{B_{n+1}}\left(\widehat{F}_n\ \mu\ \widehat{G}_n\right). \tag{116}$$

The rung residual is

$$S_{n+1} = P^{D}_{n+1} - C_{\text{close},n+1}. \tag{117}$$

If $S_{n+1} > 0$, the rung is surplus-producing; if $S_{n+1} = 0$, its scalar residual is neutral; if $S_{n+1} < 0$, it is deficit/open. Every claimed closure also requires its independent trace certificate. The sheddic surplus is

$$X_{\text{shedding},n+1} = \max(0, S_{n+1}). \tag{118}$$

5.9.2 Exoshedding lift

Exoshedding routes surplus support outside the current closure. At scope B_r, nonzero $X_{\text{exo},r}$ supplies proposed lifted support

$$K^{r+1}_{\text{prop}} = \text{Lift}_{B_r \to B_{r+1}}\left(X_{\text{exo},r}\right). \tag{119}$$

The construction trace admits it by

$$K^{r+1}_{\text{prop}} \xrightarrow{HT(K;B_{r+1})} K^{r+1}_{\text{adm}}(B_{r+1}). \tag{120}$$

Higher-scope fusion closes the admitted lifted support:

$$C^{\text{ret}}_{r+1} = \text{Close}_{B_{r+1}}\left(\widehat{F}_r \;\mu\; K^{r+1}_{\text{adm}}(B_{r+1})\right). \tag{121}$$

Thus exoshedding supplies lifted support; fusion closes it.

5.9.3 Link-closure ladder

Chain reclosure is iterated higher-scope fusion over admitted supports routed by exoshedding. A local adjacent link closure is

$$L_{i,i+1} = \text{Close}_{B_{i,i+1}}\left(\widehat{F}_i \;\mu\; \widehat{F}_{i+1}\right). \tag{122}$$

Its residual is

$$\Delta_{\text{link}} = P^{D}_{\text{link}} - C_{\text{close,link}}, \qquad X_{\text{shedding,link}} = \max(0, \Delta_{\text{link}}). \tag{123}$$

Provenance. §3.5, §5.6.5, (31).

5.10 Fractal Field Ring Gate

Scope. A fractal field ring gate is a declared recurrence motif on $B = (\partial_T F, \omega)$.

Declare

$$\mathfrak{a} \subset \mathcal{A}^{\text{ret}}_8, \qquad B = (\partial_T F, \omega). \tag{124}$$

The AF recurrence is typed here: the distinctions are declared in (125); the relations are declared in (126)–(127); the retained relation family is the FRG; $\mathfrak{S}_B$ forms the AΩ value.

5.10.1 Distinctions

Declare

$$D_{\text{FRG}} = (\mathfrak{a},\ B, V_N, I, G). \tag{125}$$

with

$$V_N = \{0, 1, \ldots, N-1\}, \qquad R_{\text{ring}} \subset F,$$

$$\partial_T R_{\text{ring}} = \text{the tesseract-admissible boundary induced on } R_{\text{ring}},$$
$$I \subseteq V_N, \qquad G = \{g_i : i \in I\}, \qquad g_i \in \partial_T R_{\text{ring}},$$
$$k \in K_{\text{bin}}.$$

5.10.2 Relations

The ring adjacency relation is

$$E_N = \{(i, i+1 \bmod N), (i, i-1 \bmod N) : i \in V_N\}. \tag{126}$$

The returned-current relation is

$$E_{\text{return}} \subseteq I \times I. \tag{127}$$

Under the adjacent-only restriction,

$$(i,j) \in E_{\text{return}} \Rightarrow (i,j) \in E_N. \tag{128}$$

Gate-to-bin and cross-return path relations are written directly as

$$\gamma : g_i \rightsquigarrow k, \qquad \gamma : g_i \leftrightarrow g_j \rightsquigarrow k.$$

The retained path-cycle carried by γ is C_γ.

Retained relation, admission and model.

$$D_{\text{FRG}} \longrightarrow R^{\text{ret}}_{\text{FRG}} \longrightarrow K_{\text{FRG,adm}}(B, i; \text{cert}) \longrightarrow (K_{\text{bin}}, \kappa, \lambda_{\mathcal{Y}}). \tag{129}$$

The retained ring adjacency and return paths carry distinction identities. After (33), choose $\kappa : K_{\text{bin}} \to \mathbb{Q}$, $\kappa_k = \kappa(k)$, and the value map $\lambda_{\mathcal{Y}}$. The sums consume paths committed through the current cut; response follows at the next transition.

5.10.3 Gate contribution

For a declared gate g_i, define the bin contribution

$$S_k^{(i)} = \sum_{t \in \omega} \sum_{\gamma : g_i \rightsquigarrow k} \kappa_k \, \mathfrak{S}_{B|t}(C_\gamma). \tag{RG1}$$

5.10.4 Independent gate-sum value

The independent gate-sum value is

$$T_k^0 = \sum_{i \in I} S_k^{(i)}. \tag{RG2}$$

For two gates,

$$T_k^0 = S_k^{(a)} + S_k^{(b)}. \tag{RG3}$$

5.10.5 Cross-return contribution

For a returned-current pair $(i, j) \in E_{\text{return}}$, define

$$C^{\times}_{ij,k}(B) = \sum_{t\in\omega} \sum_{\gamma: g_i \leftrightarrow g_j \rightsquigarrow k} \kappa_k \, \mathfrak{S}_{B|t}(C_\gamma). \tag{RG4}$$

For the two-gate specialization with $E_{\text{return}} = \{(a, b)\}$,

$$C^{\times}_{ab,k}(B) = \sum_{t\in\omega} \sum_{\gamma: g_a \leftrightarrow g_b \rightsquigarrow k} \kappa_k \, \mathfrak{S}_{B|t}(C_\gamma). \tag{RG5}$$

5.10.6 Recurrent ring value

The recurrent ring value is

$$T^{\times}_k = T^0_k + \sum_{(i,j)\in E_{\text{return}}} C^{\times}_{ij,k}(B). \tag{RG6}$$

Under the specialization of (RG5),

$$T^{\times}_k = T^0_k + C^{\times}_{ab,k}(B). \tag{RG7}$$

5.10.7 Field-ring identity

The field-ring identity is

$$T^{\times}_k - T^0_k = \sum_{(i,j)\in E_{\text{return}}} C^{\times}_{ij,k}(B). \tag{RG8}$$

Under the specialization of (RG5),

$$T^{\times}_k - T^0_k = C^{\times}_{ab,k}(B). \tag{RG9}$$

5.10.8 FRG value package

The compact FRG value package is

$$\mathcal{V}_{\text{FRG}} = (\mathfrak{a},\ B,\ R_{\text{ring}}, \partial_T R_{\text{ring}}, V_N, E_N, I, G, K_{\text{bin}}, \kappa, E_{\text{return}}, \lambda_{\mathcal{Y}}). \tag{RG11}$$

The declared value map is

$$\lambda_{\mathcal{Y}} : (T^{\times}_k)_{k\in K_{\text{bin}}} \mapsto \mathcal{Y}. \tag{RG12}$$

Remark. FRG supplies gated recurrent cross-return; fusion closes admitted support.

5.11 Field dynamics

5.11.1 Field-fluid regime

Field dynamics is the AΩD calculation of retained field, current-cycle, background-continuation, and shell-support participants on declared $B = (\partial_T F, \omega)$. A field-fluid regime is the local declaration in which those participants couple through finite SADAR capacity. For a moving or nested cell boundary, write

$$B_c|t = (\partial_T R_c(t), \omega_c|t).$$

The retained cycle family is

$$\mathcal{C}_c(t) = \mathcal{C}_{\text{field},c} \cup \mathcal{C}_{\text{current},c} \cup \mathcal{C}_{\text{bg},c} \cup \mathcal{C}_{\text{shell},c}.$$

Here

$$C_J = \text{retained current-cycle participant}, \qquad J^D = \text{accumulated current value}.$$

5.11.2 Support-resolution model

Let

$$\mathcal{C}_{a,r,\Sigma} = \text{cycle}\langle a, r\rangle(\Sigma_{\text{support}}),$$

where Σ_{support} is the declared support-expression family. The structural support multiplicity is

$$N_{\text{support}}(a, r, \Sigma; B) = \left|\text{Sec}^{(r)}_{T,\text{ret}}(a; B)\right| \left|\Sigma_{\text{support}}\right|.$$

The monitored reflection-duration accessor is selected by the declared rule

$$RD_{\text{mon}}(a, r, \Sigma; B) = \max_{C_u \in \mathcal{C}_{a,r,\Sigma}} RD_{A\leftrightarrow\Omega}(C_u; B).$$

Resolution is chosen by a declared coarseness order $\preceq_{\text{coarse}}$. Tie-breakers are declared before application. The selected q is the coarsest resolution among already admitted candidates under $\preceq_{\text{coarse}}$ after declared tie-breakers. No measured residual or score enters this gate.

5.11.3 Medium declaration and field count

Declare $N_{\text{candidate},c}, N_{\text{support},c} \in \mathbb{Z}_{\geq 0}$, $\mathcal{K}_c^{\text{raw}} \in \mathbb{Q}_{\geq 0}$, and $\bar{k}_F, \bar{\rho}_F^D \in \mathbb{Q}_{>0}$ before application. The raw maximum supported field count is

$$N_{F,c}^{\max,0} = \min\left(N_{\text{candidate},c}, N_{\text{support},c}, \left\lfloor \frac{\mathcal{K}_c^{\text{raw}}}{\bar{k}_F} \right\rfloor, \left\lfloor \frac{|\omega_c|}{\bar{\rho}_F^D} \right\rfloor\right).$$

Subject to actual recurrent member admission, the retained field family is then bounded as

$$F_{B_c} = \{F_1, \ldots, F_{n_F(c)}\}, \qquad n_F(c) \leq N_{F,c}^{\max,0}.$$

A gas/fluid classifier is a declaration on B_c, not an external material label. Chiral flip, background-continuation jitter, slosh, semi-coupling, and lead-lag are boundary-current features of the declared regime.

Remark. A field-fluid regime may distinguish locally seated, unresolved-jitter, pending, and sheddic chiral events by declared boundary status.

5.11.4 Nested boundary and inter-layer SADAR coupling

For nested boundaries

$$R_0 \subset R_1 \subset \cdots \subset R_m, \qquad B_\ell = (\partial_T R_\ell, \omega_\ell),$$

the inter-layer coupling value is

$$G^{\mathfrak{S}}_{\ell\to m}(B) = \sum_{(u,v)\in E^{\mathfrak{S}}_{\ell,m}(B)} \mathfrak{S}_B(C_u, C_v).$$

Range-prioritized coupling orders admissible $E^{\mathfrak{S}}_{\ell,m}(B)$ by the declared boundary range before summation. When a magnitude is needed, use the declared magnitude functional

$$J^{D,\mathrm{try}}_{r,c} = \sum_{(u,v)\in E_r(B_c)} \left\| \mathfrak{S}_{B_c}(C_u, C_v) \right\|_{\mathrm{decl}} .$$

5.11.5 Capacity residual and exoshedding lift

Declare the capacity residual $\Delta^{\mathrm{cap}}_{\ell} = P^{D}_{\ell} - C^{\mathrm{cap}}_{\ell}$. The separate closure residual, surplus and route partition use (107), (108) and (110) on B_ℓ. For $X_{\mathrm{exo},\ell} > 0$, apply (119), then (120) before (121). Capacity and closure compatibility retain their distinct operands.

5.11.6 FRG-mediated cross-return in nested boundaries

A ring-gated layer instantiates (125) and (129) with its own $N_\ell, I_\ell, G_\ell, K_{\mathrm{bin},\ell}, \kappa_\ell, E_{\mathrm{return},\ell}$. Its cross terms and recurrent value are (RG4) and (RG6) on B_ℓ. FRG supplies gated cross-return; the preceding lift and fusion construct the higher closure, and their iteration gives chain reclosure.

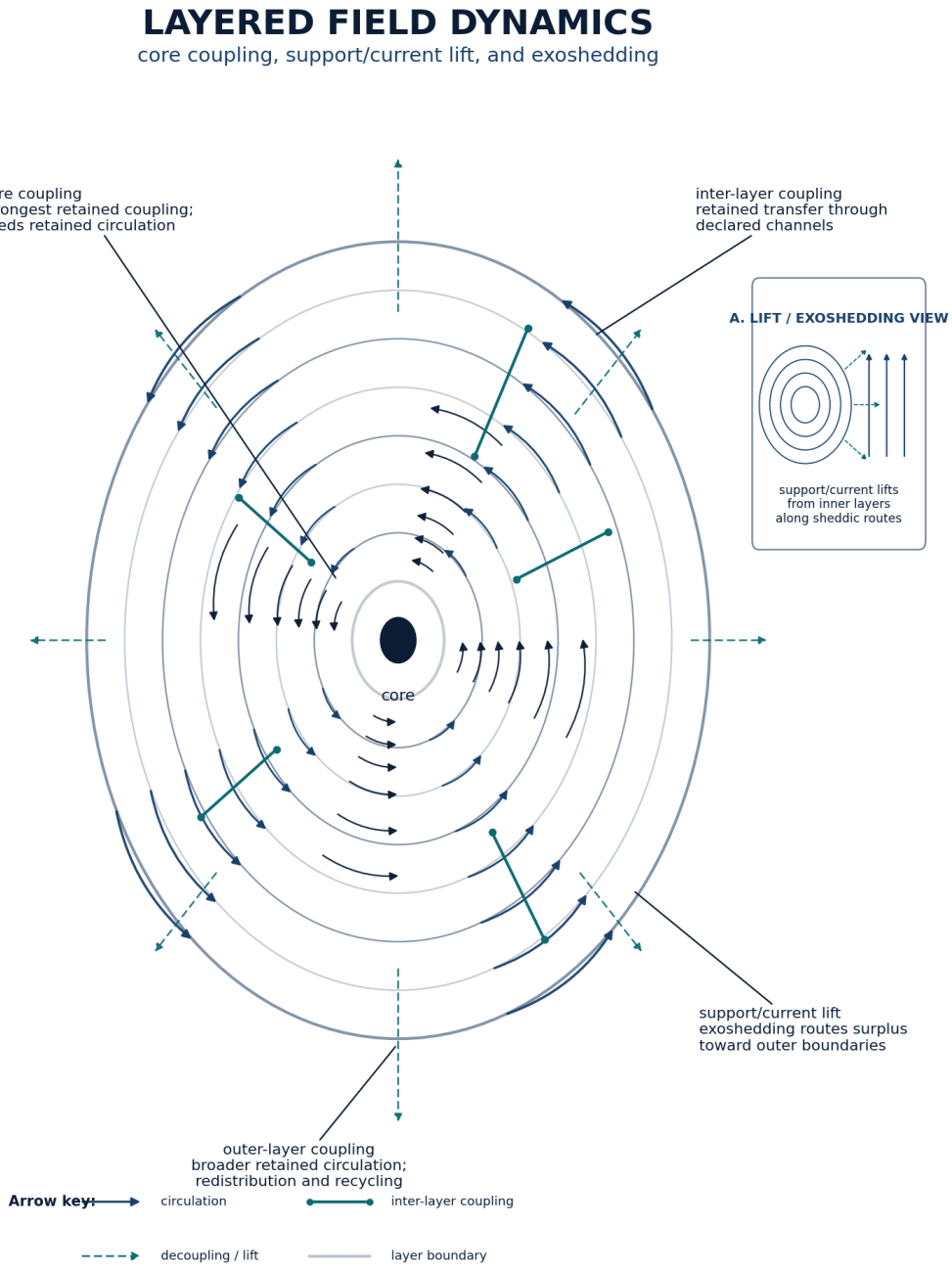


Figure 10: Layered field dynamics. Boundaries separate layers; same-sense circulation remains on each layer; coupling arrows cross declared inter-layer channels; decoupling/lift arrows indicate exoshedding after closure surplus.

5.11.7 Finite SADAR moving-boundary dictionary

On $B_c|t$, substitute $\phi_e = \mathfrak{S}(C_e)$ into (83); sum with (84) and retain the crossing/sweep/shedding split of (88). This finite dictionary expresses bobbing as signed sweep/shedding modulation on the declared moving or nested cell boundary.

Fractal-calculus spine		
Declared distinction	$\downarrow$	$x \succ x = x$ with retained hinge/trace
Declared scope	$\downarrow$	$B = (\partial_T F, \omega)$
Certified closure/recurrence	$\downarrow$	$C_f[\Sigma]_{A \leftrightarrow \Omega}$
$\mathsf{AO}_\mu(C)$	$\downarrow$	$\mathsf{AO}_\mu(C) = \langle \Pi_\alpha, H_\mu, \Pi_\Omega \rangle$
SADAR value	$\downarrow$	$\mathfrak{S}_B(C) = p^D A$
Boundary accumulation	$\downarrow$	$\Phi^D_{\partial_T R} = \sum_{e \in \partial_T R} \mathfrak{S}_B(C_e)$

Figure 11: Fractal-cycle calculus spine.

5.12 SADAR value trace of admitted recurrence

For an already admitted wave W of (24), let C_t denote its bound retained cycle on each committed cut. This closure/current contribution is the operand of (92). Its value projection is

$$v_B(W) = \{(t, \mathfrak{S}_{B|t}(C_t)) : t \in \omega\}. \tag{130}$$

The wave's admission and lineage are those of (24); this projection records their SADAR values.

The hinge μ implicitly supports chirality: it carries the handed return of the branch/hinge/branch package. The chiral cycle presentation is

$$A \leftrightarrow_\mu \Omega \to \Omega \leftrightarrow_\mu A \to A \leftrightarrow_\mu \Omega. \tag{131}$$

This presentation is part of the admitted recurring lineage: $A \leftrightarrow_\mu \Omega$ opens the returned package, $\Omega \leftrightarrow_\mu A$ carries the handed return, and $A \leftrightarrow_\mu \Omega$ records reclosure on the declared boundary/window.

SADAR value trace of admitted recurrence

C_t = retained branch/hinge/branch package on the committed cut

$A \leftrightarrow_\mu \Omega \to \Omega \leftrightarrow_\mu A \to A \leftrightarrow_\mu \Omega$ = chiral hinge return presentation

$\downarrow$ temporal SADAR presentation on ω

$v_B(W) = \{(t, \mathfrak{S}_{B|t}(C_t)) : t \in \omega\}$

Figure 12: An admitted recurring lineage has a chiral hinge-return presentation and a downstream SADAR value trace.

5.13 Fractal-tesseract cycle range

For finite $\mathfrak{a} \subset \mathcal{A}_8^{\mathrm{ret}}$, section grade r, and support expression Σ of (36), define

$$\mathcal{C}_{\mathfrak{a},r,\Sigma} = \mathrm{cycle}\langle \mathfrak{a}, r \rangle(\Sigma) = \mathcal{P}^{(r)}_{\mathfrak{a}}(\Sigma). \tag{132}$$

The fractal coordinate range selects retained addresses. The tesseract section grade selects retained sections. The support expression assigns cycle types.

The separator :: joins support clauses. If the final clause is a bare even shell number L, parse it as the shared enclosure qualifier C_L. Thus

$$3{:}4 :: 3{:}3 :: 6 \equiv (3{:}4,\ 3{:}3;\ C_6). \tag{133}$$

The structural support multiplicity is

$$N_{\text{support}} = |\operatorname{Sec}^{(r)}_{T,\text{ret}}(\mathfrak{a}; B)|\,|\Sigma_{\text{support}}|. \tag{134}$$

Tesseract sections count support occurrences. Distinct certified reclosures count beats; SADAR couples their bound recurrent lineages.

For 3:4 :: 3:3 :: 6, the support types are

$$C_\sigma[3{:}4; C_6]_{A\leftrightarrow\Omega}, \qquad C_\sigma[3{:}3; C_6]_{A\leftrightarrow\Omega},$$

with $|\Sigma_{\text{support}}| = 2$. In the declared symmetric single-hinge, neutral-reflection periodic specialization, choose recurrence periods equal to these RD accessors; then

$$RD_{A\leftrightarrow\Omega}(3{:}4) = 137, \qquad RD_{A\leftrightarrow\Omega}(3{:}3) = 61, \qquad T_{\text{sync}} = \operatorname{lcm}(137, 61) = 8357.$$

The coupling is $\mathfrak{S}_B(C_{3:4}, C_{3:3})$. Adjacency and the least common multiple alone do not establish that coupling.

Fractal-tesseract support family

$\mathfrak{a}$ fractal range + section grade r + $\Sigma = (3{:}4, 3{:}3; C_6)$

$\downarrow$

$\text{cycle}\langle \mathfrak{a}, r\rangle(\Sigma)$

$\downarrow$

$C_\sigma[3{:}4; C_6]_{A\leftrightarrow\Omega}$, $C_\sigma[3{:}3; C_6]_{A\leftrightarrow\Omega}$

$\downarrow$

tesseract sections count support occurrences; $\mathfrak{S}_B$ couples beats

Figure 13: Fractal-tesseract cycle range and support family.

The trace discipline is

$$\begin{aligned}
\Sigma/K_{\text{prop}} &\xrightarrow{\text{construction trace}} HT(K; B) \xrightarrow{\text{admission on } B} K_{\text{adm}}(B) \\
&\xrightarrow{\text{return-current trace}} RD_h(B) \xrightarrow{\bowtie_B R_h^{\text{asym}}} RCD_h(B) \xrightarrow{\text{dur}} \rho_h^D(B) \\
&\xrightarrow{\min(\cdot,|\omega_h|)} \rho^D_{\omega,h}(B) \xrightarrow{C_{3,h}} p_h^D \xrightarrow{A_h} \mathfrak{S}_B(C_h) \xrightarrow{\text{closure test}} \Delta_{\text{close}} \\
&\xrightarrow{\max(0,\cdot)} X_{\text{shedding}} \xrightarrow{\text{declared route}} \text{SheddicPath}\,.
\end{aligned} \tag{135}$$

5.14 Fractal-cycle finite accumulation

For a retained cycle family $\mathcal{C} = \mathcal{C}_{\mathfrak{a},r,\Sigma}$, let V_{beat} be the finite index of its supplied certificate-bound cycle contributions C_u. The declared coupling family $E_{\mathfrak{S}}(B)$ consists of supplied pairs of these contributions in the recurrent-lineage domain of (94). Define the finite SADAR accumulation

$$\mathcal{I}^{\mathfrak{S}}_B(\mathcal{C}) = \sum_{u\in V_{\text{beat}}} \mathfrak{S}_B(C_u) + \sum_{(u,v)\in E_{\mathfrak{S}}(B)} \mathfrak{S}_B(C_u, C_v), \tag{136}$$

This is finite SADAR accumulation of the supplied retained-cycle and coupled-lineage values; it is not continuous integration.

A fractal field ring gate is the ring specialization of finite SADAR accumulation when E_{return} is declared by returned-current ring relations.

Provenance. (73), (31), (32), (11), §4.4.

5.15 Polyrhythm coupling

A polyrhythm coupling retains the cadences of a finite family (132) on a common boundary/window. In the explicitly periodic RD-as-period specialization, its display coordinate is

$$\theta_u(t) = t \bmod RD_{A\leftrightarrow\Omega}(C_u), \qquad t \in \omega. \tag{137}$$

The display period is a model choice. On the declared $E_{\mathfrak{S}}(B)$, (94) supplies

$$\mathfrak{S}_B(C_u, C_v) = \phi^{\text{beat}}_{uv,A\leftrightarrow\Omega}(B) = p^{D}_{uv,A\leftrightarrow\Omega}(B) A_{uv,A\leftrightarrow\Omega}(B). \tag{138}$$

The polyrhythm SADAR value is the finite accumulation in (136). Ring-adjacent SADAR coupling gives the fractal field ring gate of §5.10.

Remark. A scalar duration projection of a retained cycle omits branch/hinge/branch information needed by later coupling, seat compatibility, shedding, or SADAR calculations.

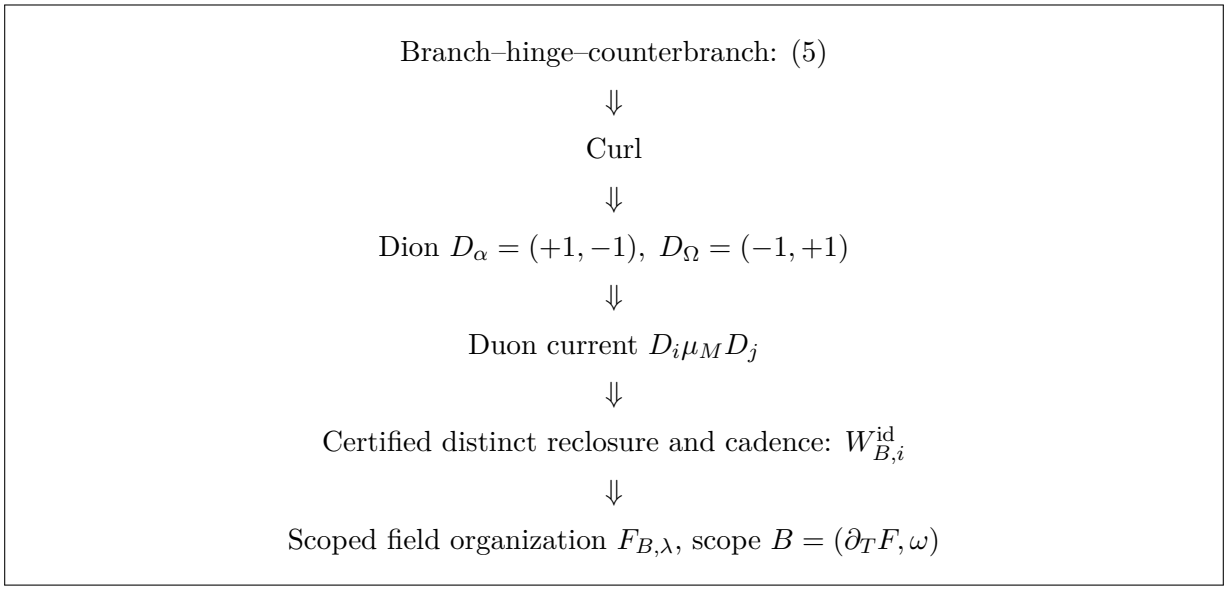


Figure 14: Curl-to-duon-current support.

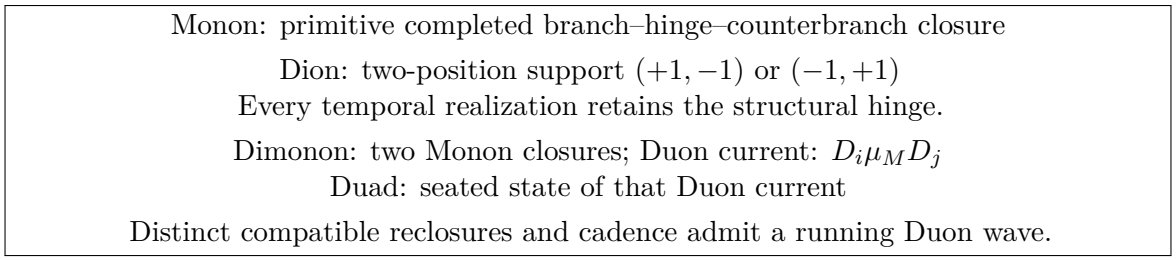


Figure 15: Primitive closure ladder.

Provenance. (4), (14), (30), App. B.

5.16 Finite harmonic coordinates of admitted recurrence

Setup. Let G be a declared finite abelian index group of order $N > 0$, and $\widehat{G}$ its exact character group. Each $\chi : G \to \mathbb{C}$ satisfies $\chi(g+h) = \chi(g)\chi(h)$ and $|\chi(g)| = 1$. Fix an admitted recurrent lineage and a value trace $v : G \to \mathbb{C}$ bound to its committed history. Translation means $T_a v(g) = v(g-a)$. Using G to index a finite record is a representation choice; periodic extension of that record supplies no proof of periodic dynamics.

5.16.1 Definition and normalization

Setup.

$$\widehat{v}(\chi) = \frac{1}{N}\sum_{g\in G} v(g)\overline{\chi(g)}, \qquad \langle v, w\rangle_G = \frac{1}{N}\sum_{g\in G} v(g)\overline{w(g)}. \tag{139}$$

All corollaries in §5.16 share this group, committed trace and normalization; local extra hypotheses are stated at their use. These value identities, including inversion, neither admit motifs/waves nor supply a response law.

5.16.2 Orthogonality

Setup. For a nontrivial character η, choose a with $\eta(a) \neq 1$.

Derivation. Translation of the finite sum gives $S = \sum_g \eta(g) = \eta(a)S$, hence $S = 0$. For the trivial character the sum is N. Thus

$$\frac{1}{N}\sum_{g\in G} \chi(g)\overline{\xi(g)} = \mathbf{1}_{\chi=\xi}, \qquad \frac{1}{N}\sum_{\chi\in\widehat{G}} \chi(g)\overline{\chi(h)} = \mathbf{1}_{g=h}. \tag{140}$$

For a cyclic factor of order L, its L characters are the powers of an exact primitive L-th root; products over a cyclic decomposition give exactly $|\widehat{G}| = N$ characters. Orthogonality therefore gives a basis of the N-dimensional function space and the second identity by the same translation argument on the dual. This establishes completeness.

5.16.3 Inversion and uniqueness

Derivation. Substitute (139) and apply the second identity:

$$\sum_{\chi} \widehat{v}(\chi)\chi(g) = \frac{1}{N}\sum_{h} v(h)\sum_{\chi} \chi(g)\overline{\chi(h)} = v(g). \tag{141}$$

Taking the inner product with any character recovers its coefficient, so no second coefficient family represents a different trace with the same full transform. Distinct provenance can still accompany equal value traces.

Falsification. A missing coefficient, altered normalization or unbound trace fails the corresponding representation or admission premise.

Provenance. (139), (140), (24), (33).

5.16.4 Linearity

Derivation. Distribute the defining finite sum:

$$\widehat{av + bw} = a\widehat{v} + b\widehat{w}. \tag{142}$$

Falsification. Applying this value identity to infer a linear closure or update law changes its domain.

Provenance. (139).

5.16.5 Shift and modulation

Setup. For $a \in G$ and $\eta \in \widehat{G}$, set $M_\eta v(g) = \eta(g)v(g)$.

Derivation. Substitution $h = g - a$ and character multiplication give

$$\widehat{T_a v}(\chi) = \overline{\chi(a)}\widehat{v}(\chi), \qquad \widehat{M_\eta v}(\chi) = \widehat{v}(\chi\overline{\eta}). \tag{143}$$

Specialization. A shift preserves coefficient magnitudes while changing phase. Equal power therefore does not identify return phase or lineage.

Falsification. A shift with the opposite translation convention requires the opposite character factor; replacing both by one untyped shift fails.

Provenance. (139).

5.16.6 Convolution and product

Setup. Use unnormalized convolution $(v * w)(g) = \sum_h v(h)w(g-h)$ and pointwise product $(vw)(g) = v(g)w(g)$ on the same finite group.

Derivation. Substituting $u = g - h$ factors the convolution sum; substituting both inversions and using orthogonality gives the product sum:

$$\widehat{v * w}(\chi) = N\widehat{v}(\chi)\widehat{w}(\chi), \qquad \widehat{vw}(\chi) = \sum_{\eta \in \widehat{G}} \widehat{v}(\eta)\widehat{w}(\chi\overline{\eta}). \tag{144}$$

Falsification. Omitting N violates the shared normalization.

Provenance. (139), (141), (140).

5.16.7 Correlation and power

Setup. Set $R_{vw}(a) = \sum_g v(g+a)\overline{w(g)}$.

Derivation. Substitute the inversions; the sum over g removes unequal characters:

$$R_{vw}(a) = N \sum_{\chi} \widehat{v}(\chi)\overline{\widehat{w}(\chi)}\chi(a), \quad \widehat{R}_{vw}(\chi) = N\widehat{v}(\chi)\overline{\widehat{w}(\chi)}. \tag{145}$$

For $v = w$, the power coefficients are $N|\widehat{v}(\chi)|^2$.

Falsification. Equal power is insufficient for equal complex coefficients, equal phase, or equal construction provenance.

Provenance. (141), (140).

5.16.8 Parseval

Derivation. At zero shift the same calculation, or direct orthogonality, gives

$$\frac{1}{N}\sum_{g} v(g)\overline{w(g)} = \sum_{\chi} \widehat{v}(\chi)\overline{\widehat{w}(\chi)}, \qquad \frac{1}{N}\sum_{g} |v(g)|^2 = \sum_{\chi} |\widehat{v}(\chi)|^2. \tag{146}$$

Falsification. Changing normalization on one side breaks equality.

Provenance. (140).

5.16.9 Real traces and standing coordinates

Derivation. For real v, complex conjugation of the defining sum gives

$$\widehat{v}(\overline{\chi}) = \overline{\widehat{v}(\chi)}. \tag{147}$$

For a non-self-conjugate pair define real coordinates

$$c_\chi = (\chi + \overline{\chi})/2, \quad s_\chi = (\chi - \overline{\chi})/(2\mathrm{i}), \qquad a\chi + \overline{a}\,\overline{\chi} = 2\,\mathrm{Re}(a)c_\chi - 2\,\mathrm{Im}(a)s_\chi. \tag{148}$$

Running character and standing pair coordinates represent the same admitted trace. Self-conjugate characters are already real and occur once.

Falsification. This coordinate change cannot classify an open word as a running wave.

Provenance. (139), (141), (147).

5.16.10 Recurrence subgroup

Setup. Let $H \leq G$, and assume the bound value trace is invariant under every translation in H.

Derivation. Shift covariance gives $(1 - \overline{\chi(h)})\widehat{v}(\chi) = 0$. Therefore

$$v(g+h) = v(g)\ (h \in H) \quad \Longleftrightarrow \quad \operatorname{supp} \widehat{v} \subseteq H^{\perp}, \quad H^{\perp} = \{\chi : \chi(h) = 1\ \forall h \in H\}. \tag{149}$$

The converse follows by translating each term in the inversion.

Specialization. Exact periodic values on a declared cyclic index may satisfy this hypothesis. Jittered cadence need not do so.

Falsification. A finite observation window alone supplies no translation invariance or exact dynamical recurrence subgroup.

Provenance. (143), (141).

5.16.11 Direct products

Setup. For $G = G_1 \times \cdots \times G_s$, characters factor as $\chi(g) = \prod_j \chi_j(g_j)$, and $N = \prod_j |G_j|$.

Derivation. If, additionally, the trace is separable, finite sums factor:

$$v(g) = \prod_j v_j(g_j) \quad \Longrightarrow \quad \widehat{v}(\chi_1, \ldots, \chi_s) = \prod_j \widehat{v}_j(\chi_j). \tag{150}$$

Specialization. Declared independent recurrence orders may use a product group. A coupled trace generally has nonseparable coefficients.

Falsification. Product index alone does not imply separability.

Provenance. (139), (140).

5.16.12 Finite cancellation residuals

For a declared region R and scope B, define

$$S_{\mathrm{int}}(R; B) = \sum_{i \in R} \sum_{j \in R} \phi_{ij}(B), \tag{151}$$

$$S_{\mathrm{div}}(R; B) = \sum_{i \in R} \mathrm{div}(i; B), \qquad S_{\partial}(R; B) = \sum_{\substack{i \in R \\ j \notin R}} \phi_{ij}(B). \tag{152}$$

A valid declaration satisfies internal cancellation,

$$S_{\mathrm{int}}(R; B) = 0, \tag{153}$$

and boundary balance,

$$\mathcal{E}(R, B) = S_{\mathrm{div}}(R; B) - S_{\partial}(R; B) = 0. \tag{154}$$

If $S_{\mathrm{int}}(R; B) \neq 0$, the declared orientation, sign, or antisymmetry fails on that scope. If $\mathcal{E}(R, B) \neq 0$, the declaration has a missing returned-current channel, incomplete boundary/window, undeclared sheddic path, or wrong channel content.

Wave slice and window projection. For an already admitted wave, the slice projection is (83) and its retained window sum is (84). These are downstream value coordinates of (24).

Comparison scope. Native counts and admitted harmonic coordinates retain their exact finite domains.

5.17 Audit hooks

Audit hooks preserve structural checks.

5.17.1 Spine audit

The audit passes when every retained occurrence at every finite depth has branch/hinge/branch incidence.

5.17.2 Boundary audit

Check the window sum (84) on its declared slices (83).

5.17.3 SADAR audit

Check the (96) operands and repeated balance of the declared (97) family on its retained topology.

5.17.4 Seat/enclosure routing audit

The audit passes when field-count scaling and seat-compatible shell/enclosure routing remain distinct calculations on the same declared boundary/window.

5.17.5 Sheddic path audit

On a declared field and window, if $X_{\mathrm{shedding}}(\partial F, \omega) > 0$ and no sheddic path is declared through E_2, C_6, C_8, C_{10}, or a higher even C_{2n} form, the shedding statement fails on that scope. A sheddic path fails on the declared scope if its retained form violates the stated $TTL_{\mathcal{C}}$ condition.

5.17.6 Symbol audit

SYM-A1–A4 check, in order, support (31), constructor (56), scalar interface (57), and compatible clipping (60).

5.17.7 Cycle and polyrhythm audit

The cycle audit is certified when a retained field cycle is declared before any scalar duration accessor:

$$\mathcal{C}_f[p{:}q] \quad \text{before} \quad RD(\mathcal{C}_f[p{:}q])\,.$$

The polyrhythm audit is certified when each coupled cycle, its boundary/window, its reflection-duration accessor, and the coupling residual $\Delta_{ab}^{\mathrm{cyc}}(B)$ are declared before the SADAR coupling term is formed.

5.17.8 Fusion closure audit

The fusion closure audit passes when admitted curling-curl support forms, hinge μ, closure scope B, branch/hinge/branch returned duration, and closure residual are declared before X_{shedding} or any lifted support is formed. Chain reclosure is audited as iterated higher-scope fusion over admitted lifted supports. FRG-mediated reclosure is audited by the declared relation family and the computed identity $T^{\times} = T^{0} + C^{\times}$.

5.17.9 Fractal field ring-gate audit

RG-A1. $\mathfrak{a} \subset \mathcal{A}_8^{\mathrm{ret}}$ is declared.

RG-A2. $B = (\partial_T F, \omega)$ is declared.

RG-A3. (V_N, E_N) is declared.

RG-A4. $\partial_T R_{\mathrm{ring}}$ is declared.

RG-A5. $I \subseteq V_N$ and $G = \{g_i : i \in I\}$ are declared.

RG-A6. κ_k is declared for every $k \in K_{\mathrm{bin}}$.

RG-A7. $E_{\mathrm{return}} \subseteq I \times I$ is declared.

RG-A8. Each path contribution uses $\mathfrak{S}_{B|t}(C_\gamma)$ or declared $\phi_\gamma(B|t)$.

RG-A9. $\lambda_{\mathcal{Y}} : (T_k^{\times})_k \mapsto \mathcal{Y}$ is declared as a value map.

RG-A10. The retained value difference obeys (RG8).

5.17.10 Naming audit

A label identifies a declared distinction, relation, motif or dynamic; its defining construction fixes its properties and scope. Measured-sector carrier names belong to applications.

5.17.11 Wave audit

For an already admitted recurrent wave, its value/state projection may be recorded as:

$$Z_f = (\mathfrak{c}_f, I_f^c, B_f^e) \to Z_{f+1} = (\mathfrak{c}_{f+1}, I_{f+1}^c, B_{f+1}^e).$$

Cycle compatibility carries wave continuity. Changed cycle compatibility routes excess burden into shedding, reclosure, pending return, or entropic duon current.

6 Operative self, learning and retained carriage

6.1 Retained rebalancing inside complete state

Setup. Resolve the admitted members of (38). For a declared operativity model, factor (52) into present coordinates X_k, immutable closure memory M_k^C, and retained prior rebalancing R_k:

$$S_k^+ = (X_k, M_k^C, R_k), \qquad P_{\mathrm{now}}(S_k^+) = X_k. \tag{155}$$

Only R_k contains operative retained rebalancing; X_k and M_k^C contain no duplicate of a removed value. The present coordinates retain identity, cut, position, fibre role, pressure, current, closure status, certificates, cadence, orientation, boundary, capacity and cycle count, bound to the admitted prefix.

Derivation. On a prior source–target contact trace of (55), a declared pressure-bearing FRG realization retains

$$r_e = \Phi_e^{\mathrm{recurrent}} - \Phi_e^{\mathrm{independent}} = \sum_{s \in I_e} c_{e,s}, \qquad R_k = \{(e, i, j, B, \ell, k_e, r_e, \Pi_e) : k_e \le k\}. \tag{156}$$

The rational oriented contributions $c_{e,s}$ are committed before retention at k_e, one transition after their input cut. Π_e binds operands, repeated contacts, both prefixes, the unwrapped comparison (63), and any carriage witness. Commitments validate supplied values; they cannot recover an absent r_e. The difference specializes (RG8) after the distinctions, relations and admission of (129).

Falsification. Missing operand, lineage, contact, phase-type or commitment invalidates retention. A residual, closure memory, persistence, phase lock or ancestry alone supplies no later causal witness.

Provenance. (52), (55), (71), (129).

Literature note. The distinction–relation–admission order follows AF and AFC [7, 8].

6.2 Channel operativity and a later response

Setup. Fix an admissible intervention cut k, the same X_k, M_k^C, admitted world, response law $\mathcal{U}$, future exogenous realization E, outcome projection O, and finite native horizon $H > 0$. In a stochastic model the same addressed random inputs are also fixed. For a provenance-bound retained relation $R_e \in \mathbf{R}_k$, structural removal is $S_k^{-e} = (X_k, M_k^C, \mathbf{R}_k \setminus \{R_e\})$; it does not insert a numeric zero record. The projection O selects realized later current, continuation, contact, or another declared outcome of $\mathcal{U}$. It excludes the retained coordinates themselves and role labels. Define channel operativity on this matched intervention by

$$\mathrm{Op}_e^{O,H}(S_k) = 1 \quad \Longleftrightarrow \quad R_e \text{ is valid} \quad \wedge \quad \exists t > k,\ t \le k + H : \quad O\big(\mathcal{U}^{t-k}(S_k; E)\big) \neq O\big(\mathcal{U}^{t-k}(S_k^{-e}; E)\big). \tag{157}$$

The operative retained-relation set in that declared context is

$$\mathcal{A}_k^{O,H} = \{R_e \in \mathbf{R}_k : \mathrm{Op}_e^{O,H}(S_k) = 1\}. \tag{158}$$

Only retention differs at the matched boundary; endogenous differences then follow the common law. Wrong lineage is inadmissible, and restoration includes value and provenance. Absence, retained zero and cancelling nonzero relations are distinct states, even when a sum extends an absent term by zero.

Falsification. A difference in current pressure, law or future exogenous input invalidates a retention-only comparison. Recounting R itself as the outcome, recovering a removed value through another channel, using an unbound phase type, or allowing a same-cut effect also fails the criterion.

Provenance. (5), (8), (54), (100).

6.3 Operative self and relational labels

Setup. Specialize (157) to the next-transition outcome ($H = 1$) and removal of each retained source coordinate R_i. Within this model, operative self names the recurrent coupled subgraph:

$$\begin{aligned} I_k^{\mathrm{op}} &= \{i : O(\mathcal{U}(S_k; E)) \neq O(\mathcal{U}(S_k \setminus R_i; E))\}, \\ V_k^{\mathrm{op}} &= \begin{cases} I_k^{\mathrm{op}} \cup \{j\}, & I_k^{\mathrm{op}} \neq \varnothing, \\ \varnothing, & I_k^{\mathrm{op}} = \varnothing, \end{cases} \\ E_k^{\mathrm{op}} &= \{(i, j) : i \in I_k^{\mathrm{op}}\}. \end{aligned} \tag{159}$$

Every edge retains its coupling witness; the target enters only with an operative source.

Derivation. For (159), the target, operative interface and other nodes instantiate, respectively, the sided roles of (5). Null has no node role.

Falsification. A phase-locked or similarly labelled node without an operative edge remains outside the subgraph (159).

Provenance. (157), (24), (81).

6.4 Persistent learning

Persistent learning is the preservation of the acquired relation through later transitions together with its later causal use.

6.5 Polycadence, folding and retained carriage

Setup. Relations may bind different cadences, signed lags and directional RCD. Crescendo is increased operative scope or depth under (157), independent of amplitude or phase lock. Availability can change participation while all retained histories survive; polyrhythm, nonzero lag and dropout use that same criterion.

Derivation. A retained effect may select a later fold offer. Apply (41) to both outer operands, including the peer: each binds its actual inner identities, certificates and scope map before outer certification. The cycle/recurrence counts and new comparison follow (22) and (64). Ordinary exposure (42) asserts no shedding. Upward carriage retains the same inner relation and fold:

$$R^{\uparrow} = (R^{\mathrm{inner}}, \mathrm{FoldWitness}, W^{\mathrm{outer}}, \Pi^{\times}_{\mathrm{outer}}), \qquad \mathrm{Close}(C^{\mathrm{outer}}) = 1 \implies \Delta\Phi(C^{\mathrm{outer}}) = 1_C. \tag{160}$$

The outer contact operands, scope, indices and value remain bound. Reapply (157) at that scope, with removal/restoration affecting a later response before its further fold offer.

Surviving identities remain available after exposure, fission or temporary loss of outer participation.

Falsification. A contact, lone outer return or recurring wave cannot replace the fold and causal witnesses of (41) and the reapplication of (157) at the outer scope of (160).

Provenance. (41), (22), (64), (42), (157).

A Fractal address, bip and biz

A.1 Scope

A retained calculation declares a fractal address only after the first finite distinction. Null, fibre zero, the encoded vertex, the anchor $\mathtt{00}_8$, and its declared scope remain distinct types.

A.2 Setup

The first anchor is co-created by the distinction in (3); it is not selected from a prior coordinate set. A later retained calculation may bind its already admitted wave lineages and comparison window by

$$\mathbf{a} \vdash (B, \{W^{\mathrm{id}}_{B,i}\}_{i\in I}, \omega).$$

Addresses below $\mathtt{00}_8$ remain out of scope and unbound.

A.3 Statement(s)

For a bound lineage, $\mathrm{bip}_i(\omega)$ counts distinct certified reclosures as defined by (21). For two admitted recurrent lineages, $\mathrm{biz}^{TF}_{i|j}(\omega)$ is defined only on the common committed repeated-contact window required by (64). Neither accessor creates the anchor, admits a motif, supplies a wave, or defines a master clock.

Provenance. (3), (8), (21), (64), §1.2.

Remark. Fractal addressing binds provenance; it does not replace the closure, recurrence, or coupling witnesses.

B Curling-curls

B.1 Curl-closure names

B.1.1 Hinge triad

The support relation and its ternary roles are (5); committed predecessor/crown/successor-horizon roles are (7).

B.1.2 Curl

Definition: §3.1.1.

B.1.3 Nested curl

Definition: §3.1.2.

B.1.4 Pin-curl

Definition: §3.1.3.

B.1.5 Curling curls

Definition: §3.1.4.

B.1.6 Monon

The primitive monon is (14).

B.1.7 Dion

Two oriented signatures: §3.2.

B.1.8 Duon

Resolved constructor: (30); twelve-variant count: §3.3.1.

B.1.9 Duad

Seating: §3.4; open/closed notation: §5.7.2 and §5.7.3.

B.1.10 Trion / tetrion / tetron / tetratrion / tetrad

$$\text{tetrion} = \text{bare four-cycle / four-pivot return closure},$$

$$\text{tetron is the } E_4 \text{ enclosure of §5.7.4},$$

$$\text{tetratrion} = 4 \times \text{trion} = 3 : 4.$$

Standalone trion is the threefold core unit. A tetrad is the seated tetron closure. Admission of each named entry resolves its scope, lineage and certificate through (33).

B.2 Curling curls

Table 1: Curling-curl family names.

Numeric form	Family name	Stem	Meaning
1 : 2	Dimonon	Dimon	$2 \times$ monon
3 : 1	Trio	Trio	$1 \times$ trion
3 : 2	Ditrion	Ditrio	$2 \times$ trion
3 : 3	Tritrion	Tritrio	$3 \times$ trion
3 : 4	Tetratrion	Tetratrio	$4 \times$ trion

B.3 Support-enclosure names

Table 2: Outer enclosure names.

Order	Name	Role
C_6	hexon	first outer enclosure
C_8	octon	lifted outer enclosure
C_{10}	decon	higher lifted outer enclosure

B.4 Count compression

B.4.1 Count basis

$$\mathcal{B}_{\text{curl}} = (3:3,\ 3:4,\ 1:2) = (\text{Tritrio},\ \text{Tetratrio},\ \text{Dimon}).$$

A count form $a : b : c :: L$ means

$$\text{Tritrio}^{a} :: \text{Tetratrio}^{b} :: \text{Dimon}^{c} :: \text{out}(L).$$

B.4.2 Basic shared hexon

$$1 : 1 : 0 :: 6 = \text{Tritrio} :: \text{Tetratrio} :: \text{hexon}.$$

Equivalent expanded numeric form:

$$(3:3)^{1} :: (3:4)^{1} :: (1:2)^{0} :: 6.$$

B.4.3 Filled shared enclosures

Table 3: Filled shared enclosure forms.

Compact form	Name	Outer enclosure
$3:3:0::6$	Tritrio^3 :: Tetratrio^3 :: hexon	hexon
$4:4:0::8$	Tritrio^4 :: Tetratrio^4 :: octon	octon
$5:5:0::10$	Tritrio^5 :: Tetratrio^5 :: decon	decon

B.5 Active naming set

Table 4: Active naming index.

Category	Entries	Anchor
Hinge/closure	H_3, Monon	(5)
Branch pattern	Dion D_α, D_Ω	§3.2
Composite closure	Dimonon $= 2\,$Monon	(34)
Returned-current support	Duon $D_i \mu_M D_j$, with bound pivot	(30)
Realized current	Carried crossings on admitted Duon support and a declared window	(84)
Seated current state	Duad	§3.4
Recurring current wave	Running Duon wave, certified recurrence and orientation	(24)
Standing organization	Bound opposite recurring lineages with declared balance	Native running and standing classifications, §2.3
Four-cycle / four-enclosure terms	tetrion, tetron E_4, tetratrion $3:4$, tetrad	App. B.1.10
Curling curls	Dimonon, Trio, Ditrion, Tritrion, Tetratrion	App. B.2
Outer enclosures	hexon, octon, decon	App. B.3
Count forms	$1:1:0::6$, filled shared enclosures	App. B.4

C Stochastic crowning

C.1 Scope

AΩD stochastic accounting is count-normalized boundary accounting over a declared field, boundary, window, and fractal address.

C.2 Setup

Declare a finite history family on the field and window:

$$\mathcal{H}_{\mathcal{F}}(\omega).$$

Each history has the form

$$h = (h_{\text{past}}, h_{\text{present}}, h_{\text{future}}).$$

C.3 Statement(s)

C.3.1 History family

The family $\mathcal{H}_{\mathcal{F}}(\omega)$ is the declared finite support for stochastic crowning on the window.

C.3.2 Crowning

The crowning map selects the present:

$$\text{Crown}(h) = h_{\text{present}}.$$

C.3.3 Alpha advantage

At one declared observation restriction, partition histories into mutually exclusive statuses: crowned-alpha and not yet returned, completed Omega return, or unresolved. A history is counted in exactly one current status; its earlier alpha event remains in provenance but is not counted again as a second current history. With this partition, define

$$N_\alpha = \#\{h : \ h \text{ has current alpha status}\},$$

$$N_\Omega = \#\{h : \ h \text{ has completed-return status}\},$$

$$N_0 = \#\{h : \ h \text{ remains unresolved}\}.$$

The total count is

$$N_{\text{tot}} = N_\alpha + N_\Omega + N_0, \qquad N_{\text{tot}} > 0.$$

The alpha advantage is

$$\Delta_\alpha(\omega) = \frac{N_\alpha - N_\Omega}{N_{\text{tot}}}. \tag{161}$$

C.3.4 Retained-history family

For channel e, the retained-window indicator is

$$\mathbf{1}_{\text{ret}}(e, \omega) = \begin{cases} 1, & \text{TTL}_e(\omega) \geq |\omega|_e, \\ 0, & \text{TTL}_e(\omega) < |\omega|_e. \end{cases}$$

The history retention indicator is

$$\mathbf{1}_{\text{ret}}(h, \omega) = 1$$

when every retained channel required by h satisfies the declared TTL policy on ω. The same mutually exclusive status partition is used after filtering. The retained-history family is

$$\mathcal{H}_{\mathcal{F}}^{\mathrm{ret}}(\omega) = \{h \in \mathcal{H}_{\mathcal{F}}(\omega) : \mathbf{1}_{\mathrm{ret}}(h, \omega) = 1\}.$$

Define

$$N_{\alpha}^{\mathrm{ret}} = \#\{h \in \mathcal{H}_{\mathcal{F}}^{\mathrm{ret}}(\omega) : h \text{ has current alpha status}\},$$
$$N_{\Omega}^{\mathrm{ret}} = \#\{h \in \mathcal{H}_{\mathcal{F}}^{\mathrm{ret}}(\omega) : h \text{ has completed-return status}\},$$
$$N_{\mathrm{tot}}^{\mathrm{ret}} = |\mathcal{H}_{\mathcal{F}}^{\mathrm{ret}}(\omega)|.$$

The retained alpha advantage is defined when $N_{\mathrm{tot}}^{\mathrm{ret}} > 0$:

$$\Delta_{\alpha}^{\mathrm{ret}}(\omega) = \frac{N_{\alpha}^{\mathrm{ret}} - N_{\Omega}^{\mathrm{ret}}}{N_{\mathrm{tot}}^{\mathrm{ret}}}. \tag{162}$$

C.4 Derivation

For each history, let $P_W(h)$ be the restriction of (106) to its temporal cycle group. The notation $\mathbb{E}_{\mathcal{H}}$ denotes finite arithmetic average over the declared history family $\mathcal{H}_{\mathcal{F}}(\omega)$:

$$\mathbb{E}_{\mathcal{H}}[P_W] = \frac{1}{N_{\mathrm{tot}}} \sum_{h \in \mathcal{H}_{\mathcal{F}}(\omega)} P_W(h).$$

C.5 Falsification

If a declared history family has $N_{\mathrm{tot}} = 0$, the alpha-advantage expression is undefined on that scope. If TTL filtering changes the retained support, retained counts must be recomputed before using $\Delta_{\alpha}^{\mathrm{ret}}$.

Provenance. §2.2, (161), (162), §5.7.12.

Remark. The probability terms are normalized counts on the declared history family.

Literature note. Finite Markov kernels: Norris [6], Levin–Peres–Wilmer [4].

D Cycle-shedding burden and reclosure model

D.1 Scope

This declared AΩD model relates lead–lag span, temporal SADAR burden, sheddic excess, outward remainder and local reclosure on one retained field. The coefficients are model parameters.

D.2 Setup

For native index t, let L_t be the lead component, G_t the lag component, S_t the disturbance span, P_t the burden, C_t local closure compatibility, X_t sheddic excess, O_t outward remainder, and R_t local reclosure.

D.3 Statement(s)

$$S_t = |L_t - G_t|, \qquad \widetilde{P}_t = \eta_{\mathrm{ret}} P_{t-1} + \gamma_{\mathrm{span}} S_t,$$
$$X_t = \max(0, \widetilde{P}_t - C_t), \qquad P_t = \widetilde{P}_t - X_t,$$
$$O_t = (1 - \lambda_{\mathrm{reclose}}) X_t, \qquad R_t = \lambda_{\mathrm{reclose}} X_t.$$

Here O_t is the non-reclosed outward remainder; route placement may later split it into exoshedding, redirected support routing, and open loss.

D.4 Falsification

A proposed burden/reclosure sequence that violates any stated recurrence for the declared parameters is not a solution of this model.

Provenance. §5.6.3, §5.6.4, §5.6.5.

E Glossary

Term	Construction in this note
Null symmetry	Undistinguished presentation; (1).
Cut-running	Running relation; (4).
Hinge triad	Sided support; crown roles in (7); (5).
Fractal tesseract	Recursive support and its finite chart; §1.4.
Four-edge x witness	Degree and ancestry; fibre remains separate; §1.4.
Fractal address	Octal address and first anchor; (3).
Fractal coordinate	The left glyph $a \mid -$, used in forms such as $0 \mid - :: Z : A : 0 : 6$. §5.7.13.
Closure	Lineage-bound return; (13).
Curl	Hinge-return form; §3.1.1.
Monon	Primitive closure; (14).
Phase-bearing step	Committed traversal count; (15).
Closure-phase cycle	Completion accessor; (16).
Accumulated closure cycles	Cumulative count; (22).
Beat	Reclosure count; window origin in (23); (21).
Cadence	Ordered separations; (21).
Wave	Admitted identity and growing history; (24).
Wave continuity	Admitted-lineage presentation; App. F.2.
Field	Scoped membership; (38).
Relational comparison	Native projection; comparator (62); (64).
Memorum	Member roles and retained organization; §3.6.
Region	Retained field part; §3.6.1.
Boundary	Retained interface; §3.6.2.
Fixed boundary/window	Count and SADAR scope; §5.2.1.
Tesseract-admissible boundary	Finite induced boundary; (11).

Term	Construction in this note
Cycle-valued field compression	Certified object; scalars are accessors; (40).
Polyrhythm coupling	Retained cadence family; §5.15.
Fractal field ring gate	Gated recurrent relation; §5.10.
Boundary gate	Declared gate incidence; §5.10.1.
Independent gate-sum value	Independent contribution sum; (RG2).
Cross-return contribution	Bound cross paths; (RG4).
Recurrent ring value	Independent plus cross value; (RG6).
Value map	Declared output map; (RG12).
Retained field family	Resolved scoped participants; §5.2.2.
Field count	Resolved participant count; (89).
Field-count term	Declared logarithmic count model; (90).
Polychiral shell dressing	Oriented outer dressing; §3.6.4.
Dion	Oriented two-position pattern; §3.2.
Duon current	Resolved support; current (84); (30).
Duad	Seating state; notation §5.7.3; §3.4.
Temporal cycle group	Current segment; §5.6.1.
Lead–lag disturbance	Asymmetric return span; §5.6.2.
Shed	Verb; §5.6.4.
Shedding	Computed surplus; (108).
Sheddic	Path/route adjective; §5.6.4.
Yro	Odd-window route status; not an enclosure; §5.6.6.
Exoshedding	Outer route placement; (110).
Endoshedding	Inner route placement; (110).
Redirected support routing	Redirected placement; (110).
Tetrion	Four-cycle name; App. B.1.10.
Tetron	Four-seat inner enclosure; §5.7.4.
Support enclosure	Outer even orders; §5.7.5.
Tetratrion	Core family name; App. B.2.
Dimonon	Two Monons; family App. B.2; (34).
Trio	Core family name; App. B.2.
Ditrion	Core family name; App. B.2.
Tritrion	Core family name; App. B.2.
Dimonhexon	Proposed enclosed specification $1:2:6$; §3.5.
Tritriohexon	Proposed enclosed specification $3:3:6$; §3.5.
Tetratriohexon	Proposed enclosed specification $3:4:6$; §3.5.
Dimonocton	Lifted name $1:2:8$; support §3.5; enclosure §5.7.7.
Dimondecon	Lifted name $1:2:10$; support §3.5; enclosure §5.7.8.
Hexon	First outer enclosure; §5.7.6.
Octon	Higher outer enclosure; §5.7.7.
Decon	Higher outer enclosure; §5.7.8.
Higher even support enclosure	Higher route orders; §5.7.9.
Tritrio::Tetratrio::hexon	Shared enclosure name; App. B.4.2.
$\bowtie_B$	Scoped reflection constructor; (56).
RCD	Directional coupled object; (56).
ρ^D	Scalar duration interface; (57).
ρ^D_ω	Compatible participation clip; TTL (113); (60).

Term	Construction in this note
Q^D	Pending overhang burden; (105).
Duonic pressure	Clipped field burden; pending (105); (91).
Entropic duon current	Retained unresolved burden; (111).
ADAR	Local returned-channel tuple; (96).
SADAR	Returned-channel family; value (92); (97).
SADAR attention vector	Pressure-bearing response input; (98).
Field symmetry	Scoped relative balance; §5.5.2.
Motion	Realized later transition; (100).
Seat compatibility	Declared reseating support; §5.7.
Crowning	Committed role advance; observation App. C.3.2; (18).
Alpha advantage	Scoped current-status statistic; (161).
Entropic carrier	Continuity pattern carrying coherent distinction-information and unresolved burden beyond cycle compatibility.
Markovian continuity	Present-state realization; §4.3.2.
Retained-history continuity	History-bearing realization; §4.3.2.
AFC flux	Normalized memory/kernel type; AOD pressure type is separate; (71).
Duon slice flux	One cut-frame value; (83).
Current	Retained window sum; (84).
Support condition	Relation-continuation capacity for a declared cycle or SADAR presentation.

Closure roles. The closure operation certifies a completed retained-start return. A compatibility value records the result of a declared comparison; mutable status records whether a continuation is pending, seated or shedding at a cut. Retained closure lineage records the immutable ancestry of completed returns. These have separate types: changing status or obtaining a compatible scalar value neither creates another return nor erases its lineage. Equations (13), (20) and (52) fix their interfaces.

Index discipline. The resolved update index k, scale ℓ, additional hinge dwell m, recurrence order L, route radius r, and primitive-order marker (0) have distinct roles. An expression that uses a different local letter declares that role at its definition. The inherited odd support-count model is

$$\nu_\ell = 2\ell + 1, \qquad \ell \in \mathbb{N}_0, \qquad \nu_0 = 1. \tag{163}$$

It counts support in that model; it supplies neither a closure-phase cycle, recurrence separation nor RCD duration. In particular, its index ℓ does not identify the atomic update count k.

F Wave presentation and audit

F.1 Scope

This appendix audits downstream presentations of an already admitted recurrent wave. Recurrence and semantic admission precede every value transform.

F.2 Cycle and wave identity

Admission and cycle/reclosure counts are (24) and (22). The presentation below consumes those witnesses.

For an already admitted wave $W_{B,i}$, C_k is its bound retained cycle from (130), with the native index k. Its SADAR-valued presentation may be written

$$\mathrm{Present}_B(W_{B,i}) = \{(k, \mathfrak{S}_{B|k}(C_k)) : k \in \omega\}. \tag{164}$$

F.3 Continuation and direction

Use the advance (18) and the native running and standing classifications in §2.3, retaining each contributing identity and orientation.

F.4 Flux and window current

The slice flux (83) and retained window current (84) are downstream value coordinates of the admitted wave. Aggregate cancellation preserves both oriented channel histories.

F.5 Falsification

The audit fails if a presentation is used without an admitted recurring wave, declared boundary/window, committed values, lineage provenance, or the required closure and reclosure witnesses. Exact Fourier inversion of an unbound word does not repair any missing witness.

Provenance. (16), (21), (24), (130), (83), (84), (139).

G Scoped flux cancellation

G.1 Scope

This App records symbolic support for finite Stokes with separately typed AΩD pressure-resolved flux values. The scope is a finite declared region R and boundary/window $B = (\partial F, \omega)$.

G.2 Statement(s)

$$\phi_{ij} \mapsto \phi_{ij}(B), \qquad \phi_{ij,A\leftrightarrow\Omega}(B) = p^D_{ij,A\leftrightarrow\Omega}(B) A_{ij,A\leftrightarrow\Omega}(B).$$

The scoped boundary pairing is retained through the AFC Stokes identity:

$$\sum_{i\in R} \mathrm{div}(i;B) = \sum_{\substack{i\in R\\ j\notin R}} \phi_{ij}(B). \tag{165}$$

The Stokes residual is

$$\mathcal{E}(R,B) = \sum_{i\in R} \mathrm{div}(i;B) - \sum_{\substack{i\in R\\ j\notin R}} \phi_{ij}(B). \tag{166}$$

A valid declared scope has $\mathcal{E}(R,B) = 0$.

G.3 Falsification

If the symbolic residual is nonzero, the declared scope has a missing channel, sign error, incomplete boundary, or undeclared sheddic path.

Provenance. (79), (81), (72), (82), (154).

H Reflection Duration on Q_4

H.1 Scope

This App states graph support, ternary fibre, Markov kernels, path histories, RD/RCD and integer support reductions for AΩ-internal SADAR pressure.

H.2 Setup

Use (8) for the finite graph and (5) for the distinct ternary fibre $\sigma_t(e) \in \{-1, 0, +1\}$.

H.3 Choice: Markov kernel

The downstream kernel (66) is normalized only for positive total weight on admitted H1 support. Equal weights give the isotropic $1/N_B$; unequal weights, including dependence on committed lag, orientation and reflection, give its anisotropic specialization. Outside that support its probability is zero. Additional dwell uses the separate fibre family (65); an empty positive-weight support supplies no normalized kernel.

H.4 Statement(s)

A path history is

$$h = ((e_0, \sigma_0), \ldots, (e_m, \sigma_m)).$$

Equation (31) specializes to this history:

$$RD_h(B) = \Pi_h^-(B) + H_{\mu,h}(B) + \Pi_h^+(B). \tag{167}$$

If the window contains each support occurrence exactly once, the full-trace specialization is

$$RD_h(B) = \sum_{t \in \omega} \mathbf{1}[h \text{ retained, pending, or returning at } t].$$

Outside that full-trace specialization the indicator sum is window participation, not full RD. Couple with (56) and project duration by (57). In a common ordered count domain, (60) clips participation, while (104) and (105) record pending excess.

H.5 Walk support corollaries

The declared model (35) has the walk accounting

$$\Pi(p{:}q) = q^p + q. \tag{168}$$

Here q^p is the memoryless q-direction walk count and q is the retained Hamming–1 edge-shell term. The branch/hinge/branch reflected-return support is

$$RD(p{:}q; B) = 2\Pi(p{:}q) + H_\mu(B) \tag{169}$$

when the branch supports are symmetric. The single-hinge reduction is

$$H_\mu(B) = 1 \quad \Longrightarrow \quad RD(p{:}q) = 2\Pi(p{:}q) + 1. \tag{170}$$

For 3:4, under the single-hinge reduction,

$$\Pi(3{:}4) = 4^3 + 4 = 68, \qquad RD(3{:}4) = 2 \cdot 68 + 1 = 137. \tag{171}$$

The added 4 is retained edge-shell support. Symmetry invokes (32); the unit hinge is an extra hypothesis, not the general rule.

H.6 Integer support reductions

The declared integer reductions are

$$1{:}2{:}6, \quad 1{:}2{:}7, \quad 1{:}2{:}8, \quad 1{:}2{:}9, \quad 1{:}2{:}10, \quad 3{:}3{:}6, \quad 3{:}4{:}6.$$

Remark. The cited combinatorics is introduced at the first Q_4 / Hamming–1 derivation in §1.4. This App uses that support to state AΩD's retained edge-shell term $+q$ and branch/hinge/branch reflected-return closure $\Pi^- + H_\mu + \Pi^+$. The ternary hinge fibre is carried on the finite tesseract witness; it is not identical with the tesseract support.

Literature note. Dyck/Catalan path combinatorics: Stanley [9], Flajolet–Sedgewick [2].
Markov kernels and random walks: Norris [6], Levin–Peres–Wilmer [4], Lawler–Limic [3].

H.7 Walk-support audit

A walk-support reduction records declared first-branch support, hinge recurrence, reflected return-branch support, the retained edge-shell term, and the AΩD branch/hinge/branch closure. The single-hinge accessor $2\Pi + 1$ is used when the single-hinge symmetric reduction is declared. A kernel reduction records a normalized probability family over admissible Hamming–1 continuations.

Provenance. §1.4, (11), (31), (32), App. H.5.

I Path Complexity and 4-Step Closure

I.1 Scope

This declared model specifies integer path complexity and a four-step load convention on finite path histories.

I.2 Statement(s)

Write an outer index as

$$L = 4c_L + r_L, \qquad r_L \in \{0, 1, 2, 3\}.$$

For a path history h, the 4-step closure load is

$$\Lambda_4(h; B) = c_h(B).$$

For a field family,

$$\Lambda_4(F; B) = \sum_{h \in \mathcal{H}_F(B)} C_{3,h} c_h(B). \tag{172}$$

I.3 Path-growth reference

The raw ternary assignment space is

$$3^n.$$

The AFC path-growth bound is recorded as

$$O(n \log n).$$

The four-step decomposition is a declared load convention, not a restriction on all closure lengths. The AFC path-growth statement concerns its declared compressed information bound, not the raw number of ternary words. These statements concern finite path histories and the declared compressed information bound.

I.4 Falsification

A declared 4-step closure reduction is contradicted on a scope when its $L = 4c_L + r_L$ decomposition is not integer-valued, its history weights are undeclared, or its compressed value is not traceable to $\Lambda_4(F; B)$.

Provenance. App. H, (75), (76).

Literature note. AFC scale-invariant branching and compressed path-information bound: Poyau [8, §3.1; Choice 6; Consequence 3], source revision `5a25e8bd5bbc`.

J A↔ Ω Field Fractal Properties

J.1 Scope

This declared model records AΩ field-support accessors and field-property invariants. The retained cycle package is written

$$C_f[M]_{A \leftrightarrow \Omega}.$$

The accessors below are defined at the declared AΩ field-support level.

J.2 Branch/hinge/branch accessor

Use the branch-order object (44), its cardinality accessor (49), and the hypotheses of (50). Under the declared single-hinge neutral-reflection field-property reduction, $\rho^D = RD_{A\leftrightarrow\Omega}$. A compatible native count window then supplies (60), channel pressure (77), and pending burden (105). These are scalar accessors of $C_f[M]_{A\leftrightarrow\Omega}$, with no change to its certified object.

Provenance. §4.4, (31), (60), (105), (108).

Remark. This declared model records AΩ field-support accessors only.

References

[1] Andrew Duncan. Powers of the adjacency matrix and the walk matrix. *The Collection*, 9:4–11, 2004. Institutional repository: `https://www.um.edu.mt/library/oar/handle/123456789/24439`.

[2] Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009. Book reference.

[3] Gregory F. Lawler and Vlada Limic. *Random Walk: A Modern Introduction*. Cambridge University Press, 2010. DOI: `https://doi.org/10.1017/CBO9780511750854`.

[4] David A. Levin, Yuval Peres, and Elizabeth L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society, 2009. First edition reference.

[5] MIT OpenCourseWare. Mathematics for Computer Science: Graph Theory / adjacency matrix walks. 6.042J course materials, `https://ocw.mit.edu/courses/6-042j-mathematics-for-computer-science-spring-2015/`, 2015. Accessed 2026-06-09.

[6] J. R. Norris. *Markov Chains*. Cambridge University Press, 1997.

[7] Reginald Poyau. Axiomatic Fundamentalism (AF): A Logical Protocol for Traceable Research (all versions), 2025. All versions. DOI: `https://doi.org/10.5281/zenodo.17561186`.

[8] Reginald Poyau. Axiomatic Fundamentalism Calculus (AFC) (all versions), 2025. All versions. DOI: `https://doi.org/10.5281/zenodo.17795590`.

[9] Richard P. Stanley. *Catalan Numbers*. Cambridge University Press, 2015. DOI: `https://doi.org/10.1017/CBO9781139871495`.

[10] Eric W. Weisstein. Hypercube Graph. Wolfram MathWorld, `https://mathworld.wolfram.com/HypercubeGraph.html`, n.d. Accessed 2026-06-09.